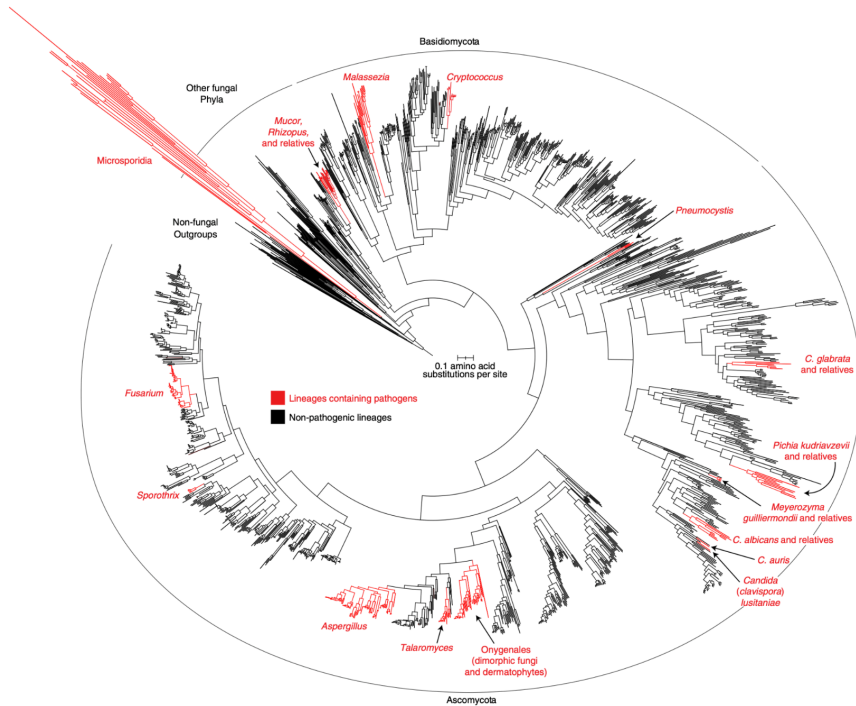


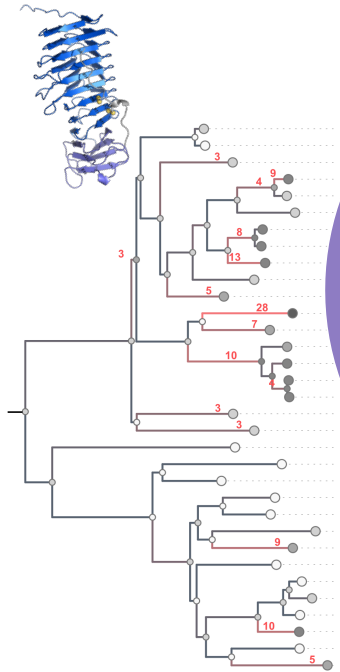
ABOUT ME

- 上海交大附中 (99-02)
- 北京大学生命科学专业 (02-06, 本科毕设: 微流系统应用于细菌耐药性)
- 美国芝加哥大学演化与生态系 (06-12, PhD thesis: evolution of gene regulatory networks and fly model of human disease)
- 美国哈佛大学系统生物学中心 (12-17, Postdoc: evolution of stress response regulation in diverse yeasts)
- 美国爱荷华大学生物系 (18-now, independent lab: evolution of stress response and adhesin gene families in fungal pathogens)
- <https://binhe.org>
- <https://binhe-lab.org>
- bin-he@uiowa.edu

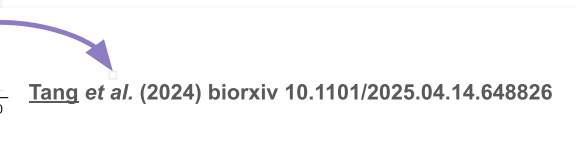
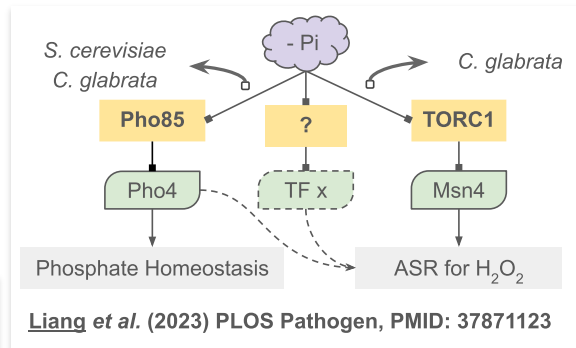
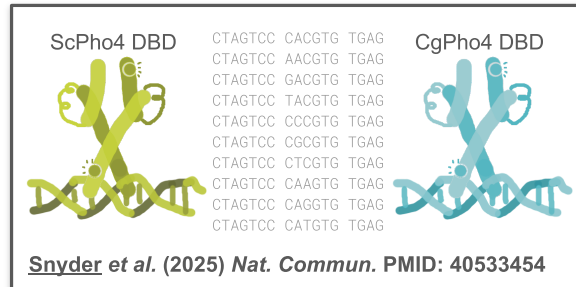
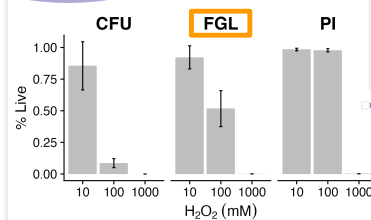
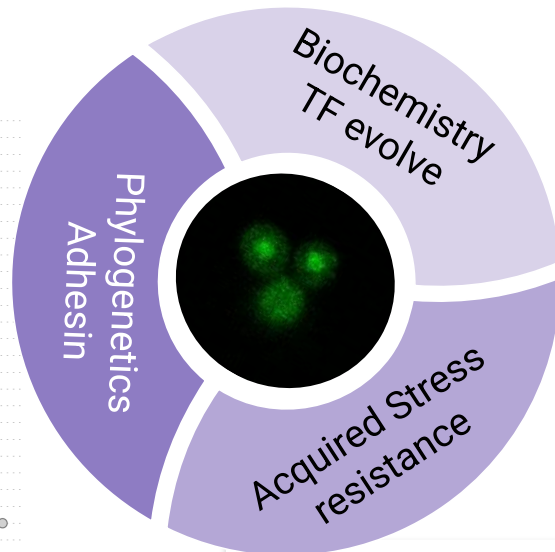
Multiple origins of pathogenic ability

- *Different mechanisms?*
- *Shared (convergent) genomic changes?*





Smoak and Snyder *et al.* (2023)
Genetics, PMID: 36794645



INTRODUCTION

WHAT IS POPULATION GENETICS? WHAT DOES IT STUDY?

POPULATION GENETICS

- A **population** is a group of individuals from the same species that lives in the same geographic area and that actually or potentially interbreeds.

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- Allele frequency: frequency of an allele in a population.
- Population genetics studies the genetic variation in populations and how it changes over time

WHY STUDY POPULATION GENETICS?

Biological evolution is the change over time in the **genetic composition of a population**, which changes due to the birth and death of individuals or the migration of individuals in or out of the population.

WHAT ARE THE POPULATION EVENTS DRIVING BIOLOGICAL EVOLUTION?

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- birth

WHAT ARE THE POPULATION EVENTS DRIVING BIOLOGICAL EVOLUTION?

- birth
- death

WHAT ARE THE POPULATION EVENTS DRIVING BIOLOGICAL EVOLUTION?

- birth
- death
- migration

WHAT ARE THE FORCES DRIVING BIOLOGICAL EVOLUTION?

- mutation
- recombination
- natural selection
- migration
- (random/sampling) genetic drift

TYPES OF GENETIC VARIATION

- SNP

TYPES OF GENETIC VARIATION

- SNP
- microsatellites

TYPES OF GENETIC VARIATION

- SNP
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 - What are microsatellites?

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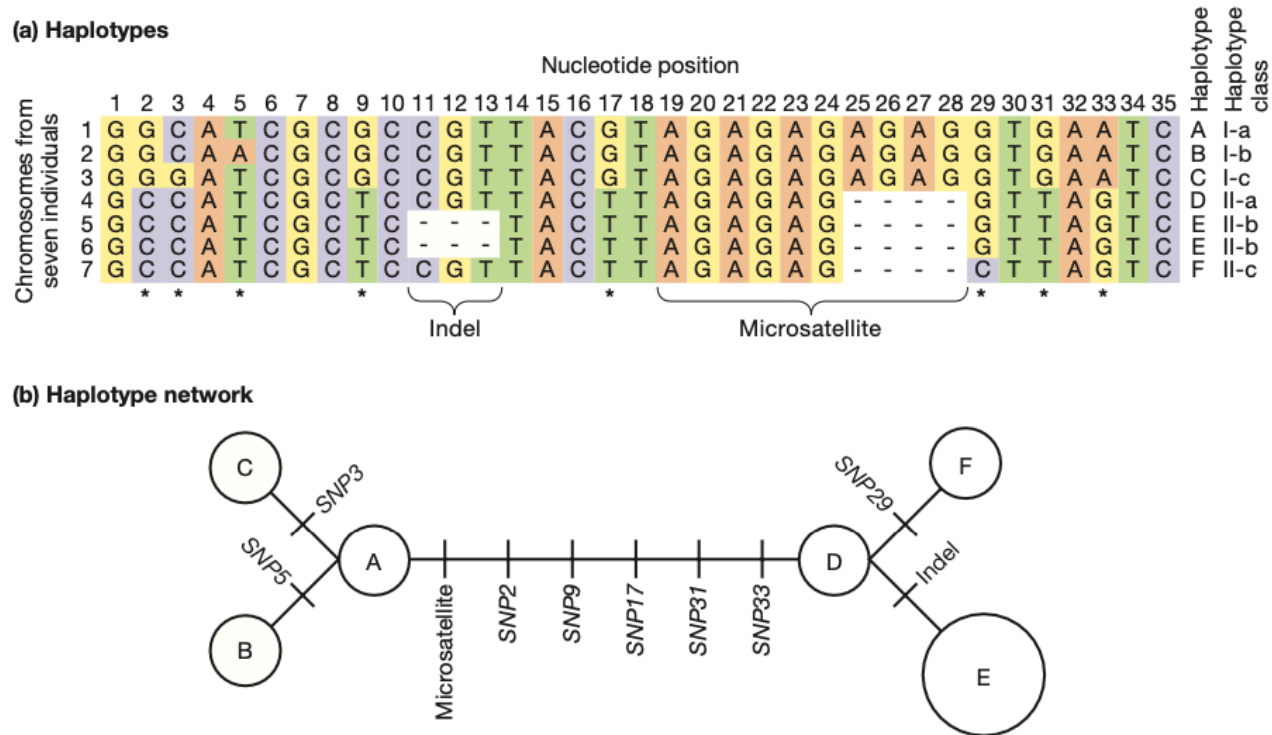
TYPES OF GENETIC VARIATION

Haplotype

refer to the combination of alleles at multiple loci on the same chromosomal homolog.

HOW DO YOU DRAW THE NETWORK (B) BASED ON THE ALIGNMENT (A)?

A haplotype network shows the relationship among haplotypes



1. Introduction to Genetic Analysis, ed 11, Fig. 18-4

A prevalent Y-chromosome haplotype among Asian men
may trace back to Genghis Khan

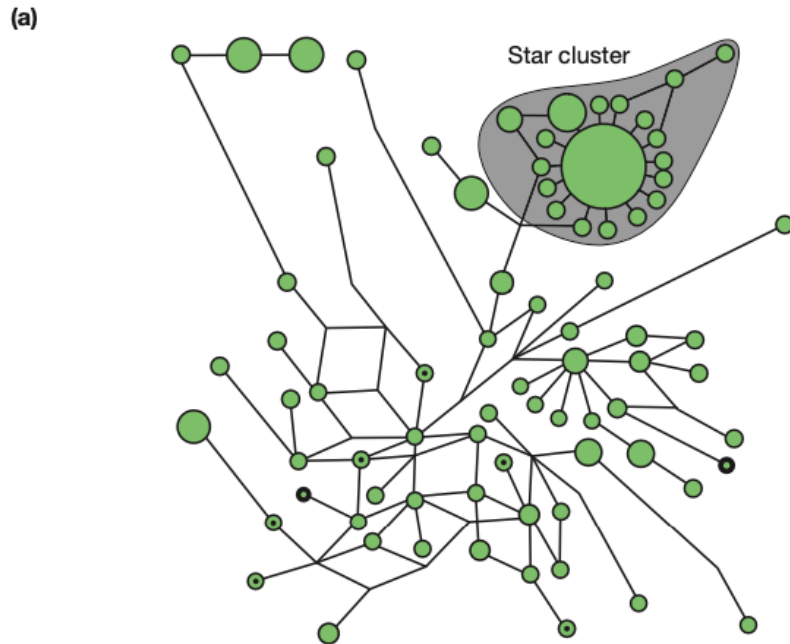
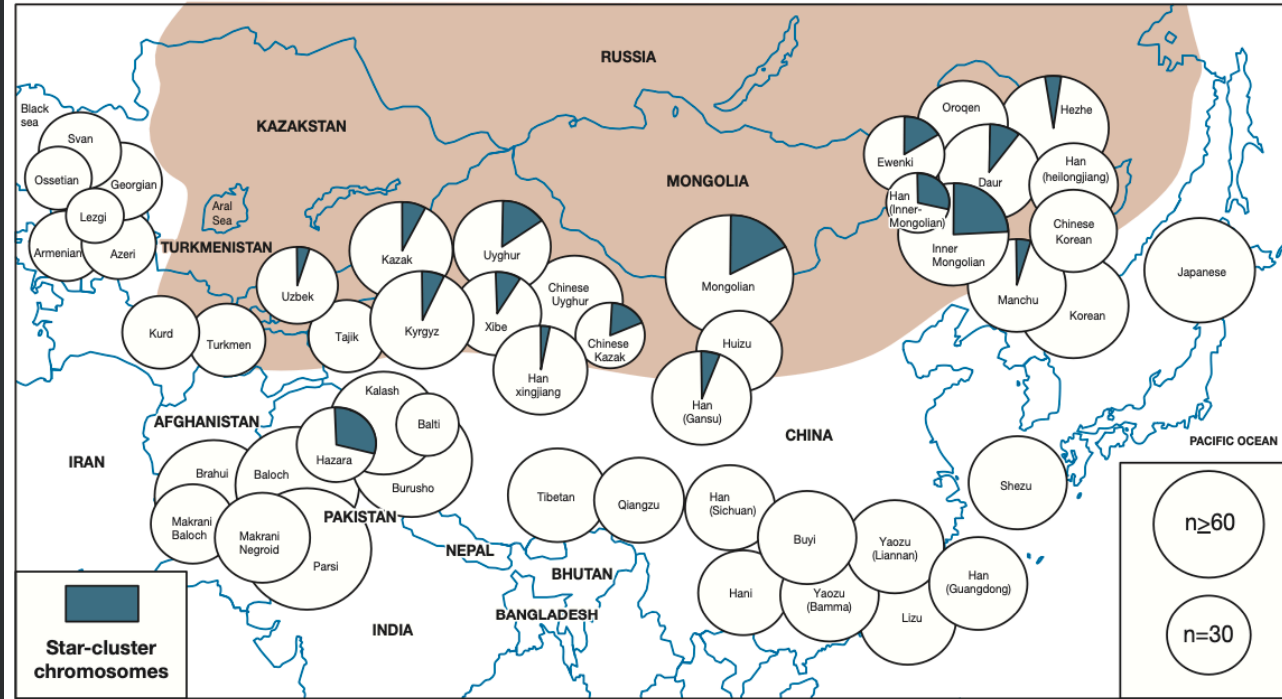


FIGURE 18-5 (a) Haplotype network for the Y chromosomes of Asian men showing the predominance of the star-cluster haplotype thought to trace back to Genghis Khan. The area of the circle is proportional to the number of individuals with the specific haplotype that the circle represents. (b) Geographical distribution of the star-cluster haplotype. Populations are shown as circles with an area proportional to sample size; the proportion of individuals in the sample carrying star-cluster chromosomes is indicated by green sectors. No star-cluster chromosomes were found in populations having no green sector in the circle. The shaded area represents the extent of Genghis Khan's empire. [Data from T. Zerjal *et al.*, *Am. J. Hum. Genet.* 72, 2003, 717–721.]

DISTRIBUTION OF THE STAR-CLUSTER HAPLOTYPE

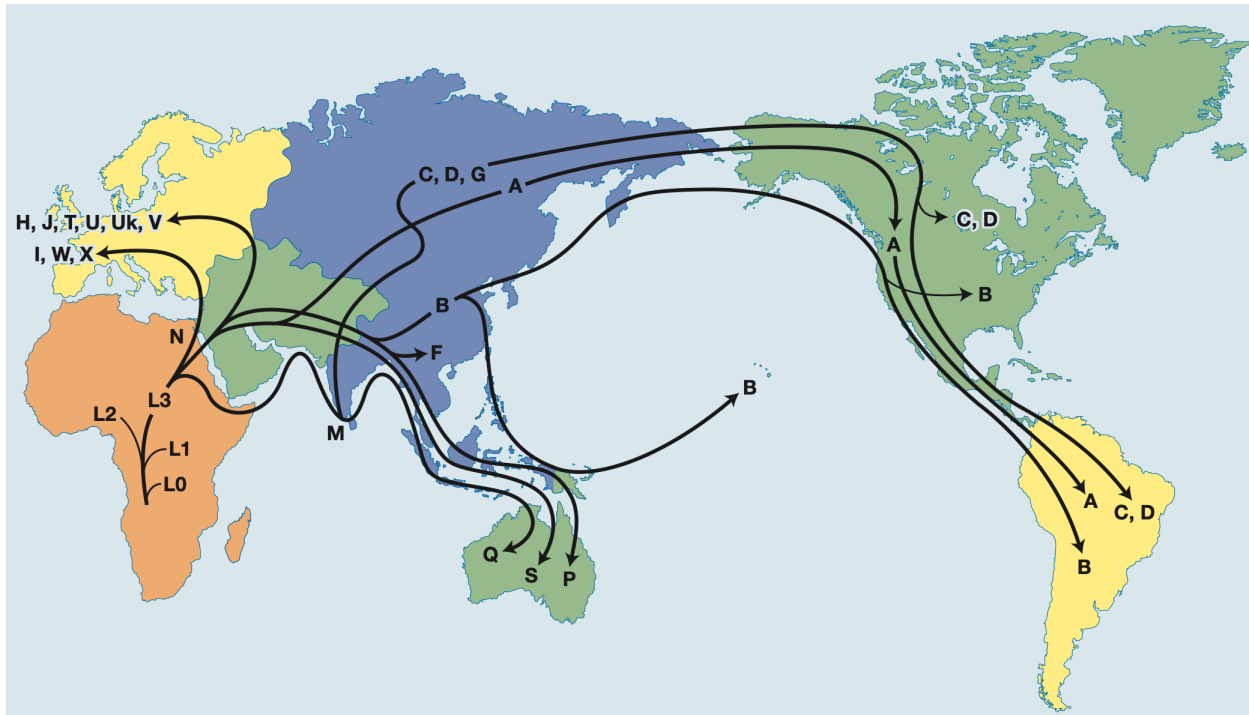
(b)



1: Introduction to Genetic Analysis, ed 11, Fig. 18-5(b)

WHERE DO WE COME FROM?

Mitochondrial haplotypes can be used to trace human origins to Africa



1: Introduction to Genetic Analysis, ed 11, Fig. 18-6

GENE POOL

- We have been evaluating genetic variations from the individual's perspective ("horizontally" as a haplotype)

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- We can also look from a site / locus / gene perspective ("vertically").

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- We have been evaluating genetic variations from the individual's perspective ("horizontally" as a haplotype)
- We can also look from a site / locus / gene perspective ("vertically").
- Introduce the **gene pool** concept

GENE POOL

Gene pool

Sum total of all alleles in the breeding members of a population at a given time.

GENE POOL AND ALLELE FREQUENCIES

A haplotype network shows the relationship among haplotypes

(a) Haplotypes

Chromosomes from seven individuals	Nucleotide position																																			Haplot	Haplot class						
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35								
	1	G	G	C	A	T	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	A	I-a					
	2	G	G	C	A	A	C	G	C	C	C	G	T	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	B	I-b					
	3	G	G	G	A	T	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	C	I-c					
	4	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	A	G	-	-	-	G	T	T	A	G	T	C	D	II-a				
	5	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	A	G	-	-	-	G	T	T	A	G	T	C	E	II-b				
	6	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	A	G	-	-	-	G	T	T	A	G	T	C	E	II-b				
	7	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	A	G	-	-	-	C	T	T	A	G	T	C	F	II-c				
			*	*		*			*			Indel					*		Microsatellite											*		*											

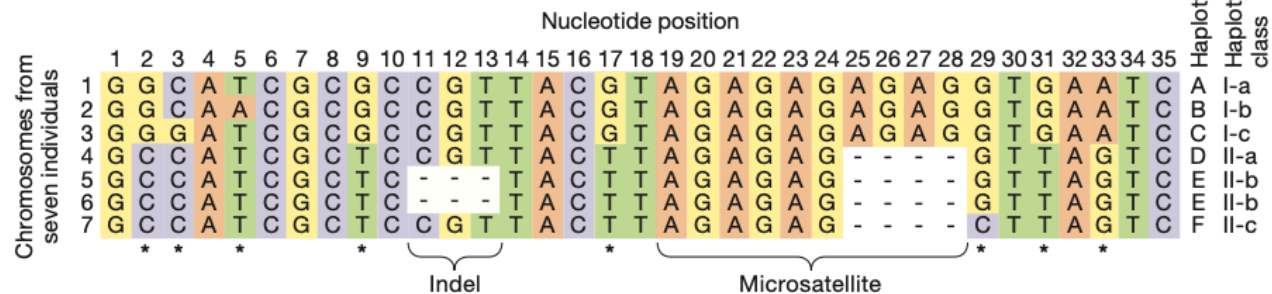
Indel

Microsatellite

GENE POOL AND ALLELE FREQUENCIES

A haplotype network shows the relationship among haplotypes

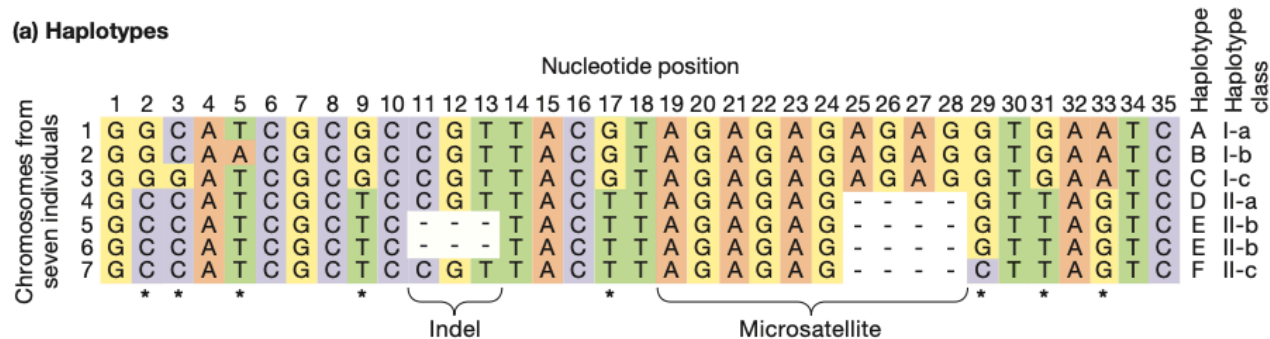
(a) Haplotypes



- **Locus**: "region of interest", may be an entire gene, or a single nucleotide

GENE POOL AND ALLELE FREQUENCIES

A haplotype network shows the relationship among haplotypes

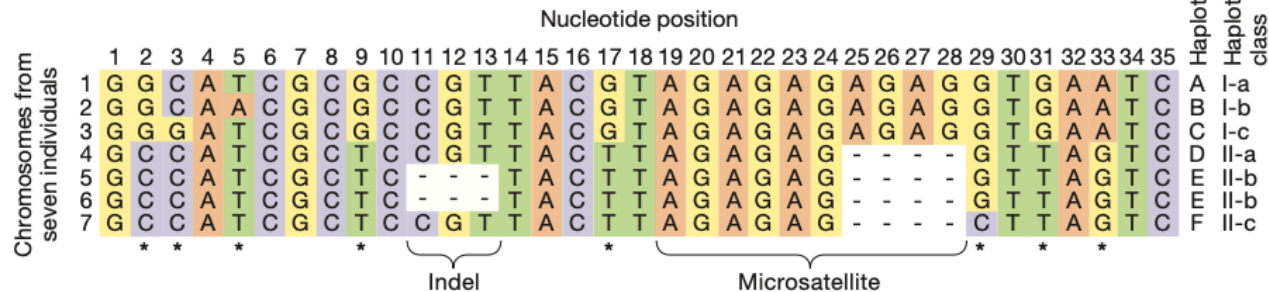


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- **Allele**: different "versions" of the sequence existing in a population

GENE POOL AND ALLELE FREQUENCIES

A haplotype network shows the relationship among haplotypes

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- How many alleles are there if we take the entire haplotype region as a locus? what's the frequency of allele E (II-b)?

GENE POOL AND ALLELE FREQUENCIES

A haplotype network shows the relationship among haplotypes

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Chromosomes from seven individuals	Nucleotide position																																			Haplotype	Haplotype class			
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	1	G	G	C	A	T	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	A	I-a		
	2	G	G	C	A	A	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	B	I-b		
	3	G	G	G	A	T	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	C	I-c		
	4	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	A	G	A	G	-	-	-	-	A	A	T	C	D	II-a	
	5	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	E	II-b
	6	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	F	II-b
	7	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	A	G	-	-	-	-	C	T	T	A	G	T	C	F	II-c
			*	*		*			*									*												*		*		*						
														</																										

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- How about just looking at site 9? What's the frequency of G?

GENE POOL AND GENOTYPE FREQUENCIES

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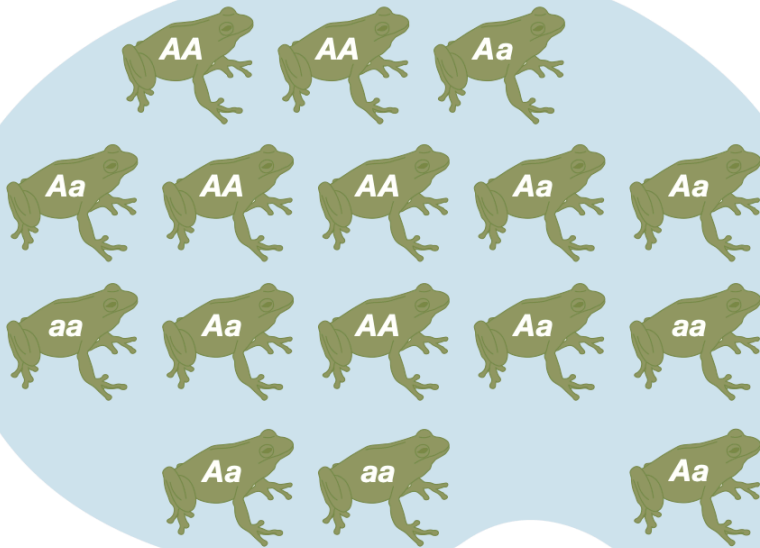
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GENE POOL AND GENOTYPE FREQUENCIES

The gene pool is the sum total of alleles in a population

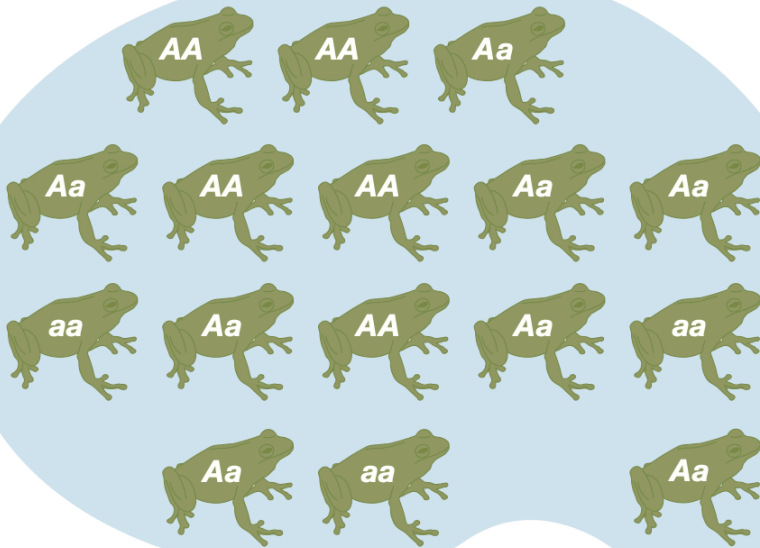


Genotypes	AA	Aa	aa
Number	5	8	3
Alleles	A	a	
Number	18	14	

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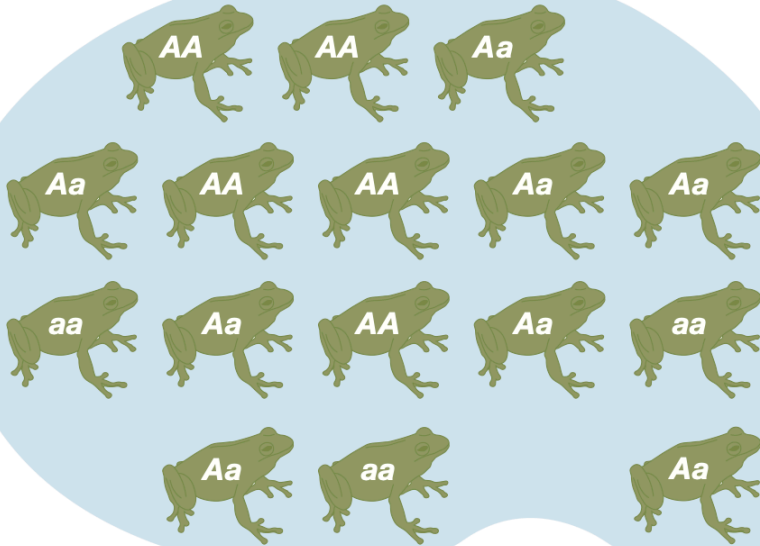


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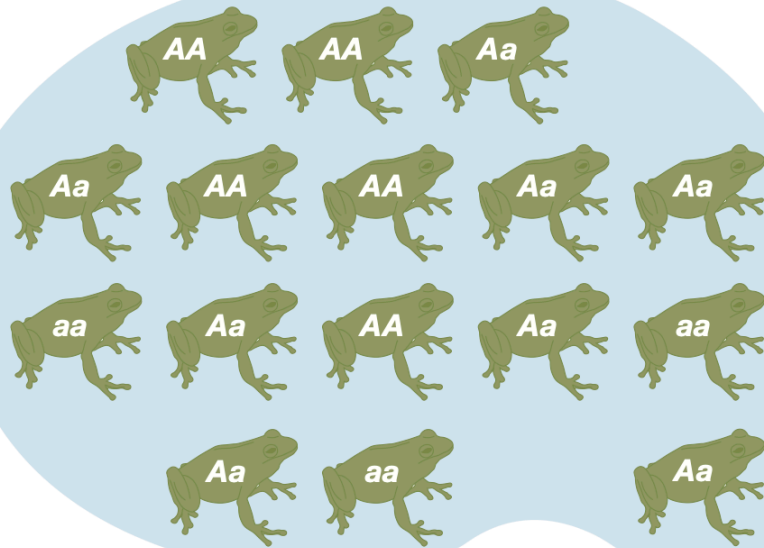
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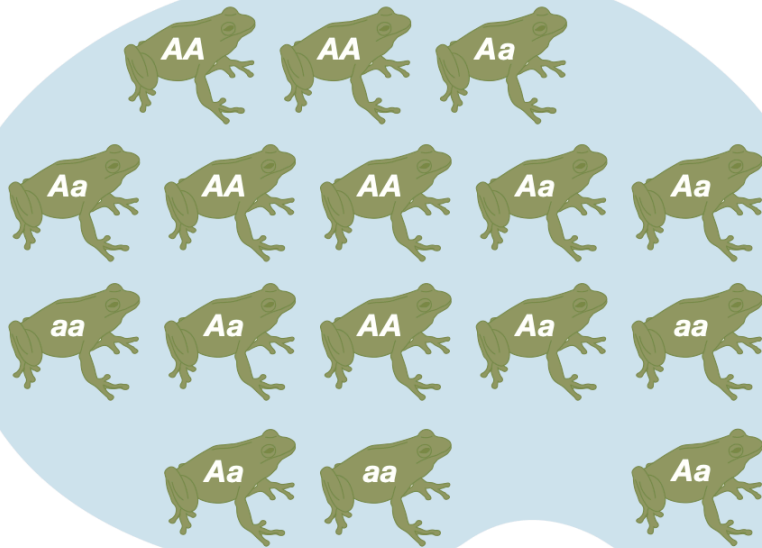


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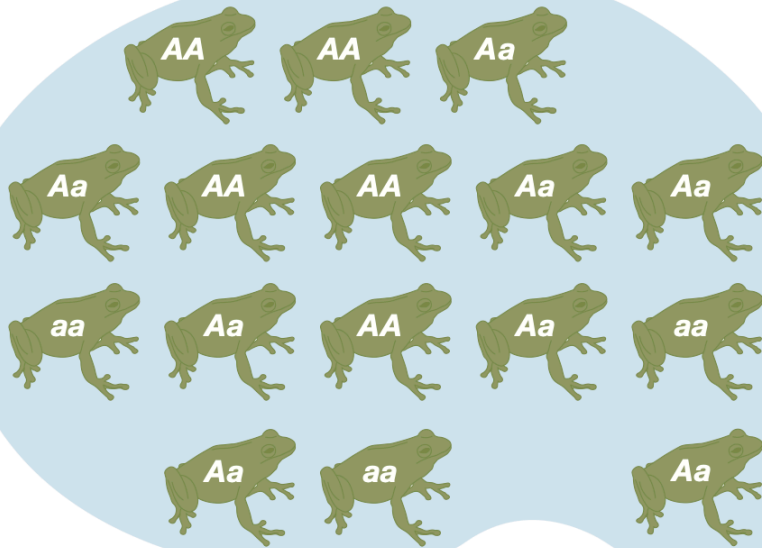


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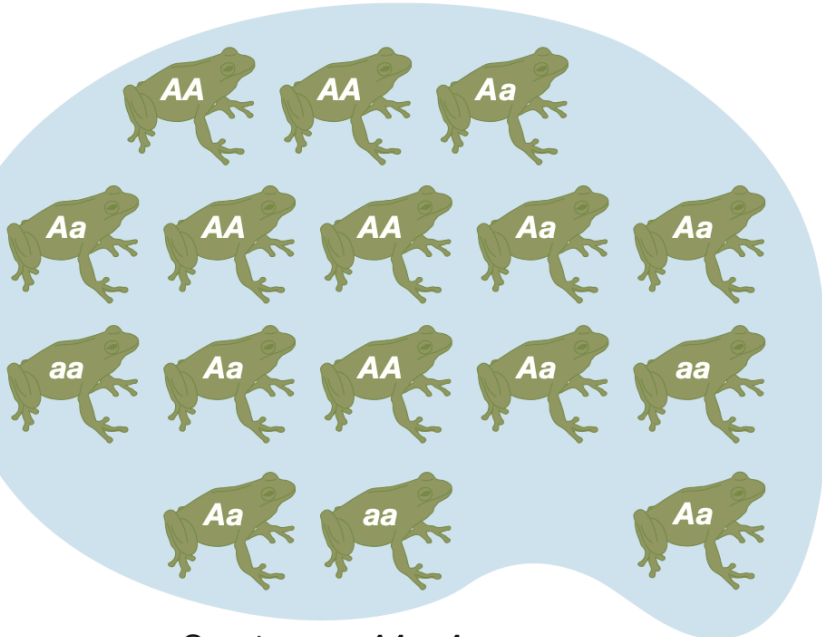
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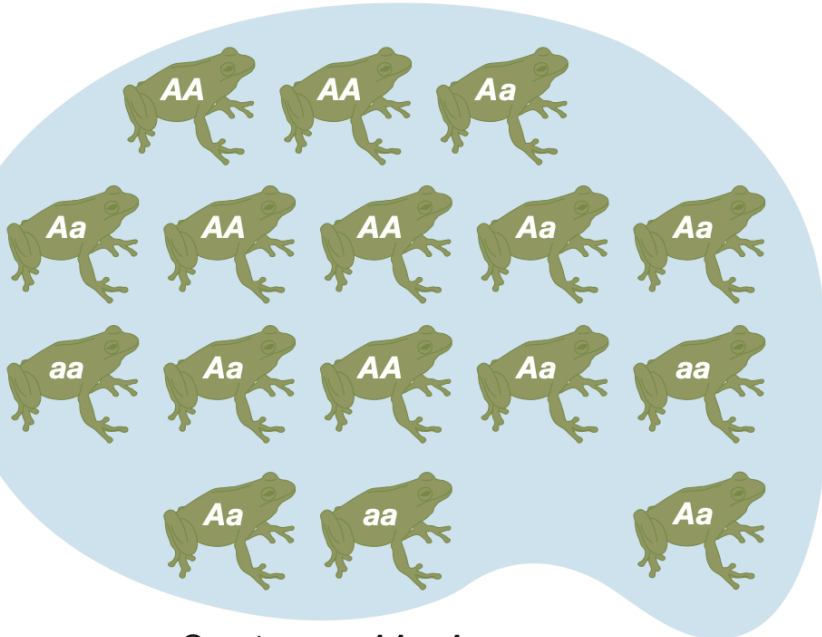
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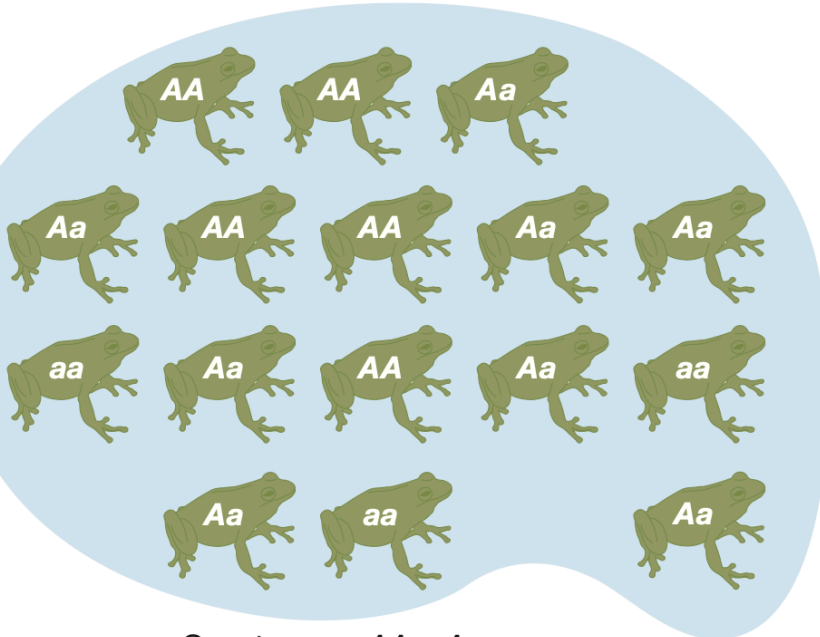


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Alleles	A	a	
Number	18	14	

- Knowing the allele frequency ($p_A = 18/32$), what is a "reasonable guess" of the genotype frequency (e.g., for $f_{A/A}$)?
- How close is it to the observed genotype frequency?
- Now reflect on your intuitive guess, what assumptions need to be made for it to work?

HARDY-WEINBERG LAW

HISTORY OF HARDY-WEINBERG LAW

JULY 10, 1908

SCIENCE

N. S. Vol. XXVIII: 49-50

DISCUSSION AND CORRESPONDENCE

Mendelian Proportions in a Mixed Population

To The Editor of Science: I am reluctant to intrude in a discussion concerning matters of which I have no expert knowledge, and I should have expected the very simple point which I wish to make to have been familiar to biologists. However, some remarks of Mr. Udry Yule, to which Mr. R. C. Punnett has called my attention, suggest that it may still be worth making.

In the *Proceedings of the Royal Society of Medicine* (Vol. I., p. 165) Mr. Yule is reported to have suggested, as a criticism of the Mendelian position, that if brachydactyly is dominant "in the course of time one would expect, in the absence of counteracting factors, to get three brachydactylous persons to one normal."

It is not difficult to prove, however, that such an expectation would be quite groundless. Suppose that Aa is a pair of Mendelian characters, A being dominant, and that in any given generation the numbers of pure dominants (AA), heterozygotes (Aa), and pure recessives (aa) are as $p:2q:r$. Finally, suppose that the numbers are fairly large, so that the mating may be regarded as random, that the sexes are evenly distributed among the three varieties, and that all are equally fertile. A little mathematics of the multiplication-table type is enough to show that in the next generation the numbers will be as

$$(p+q)^2 : 2(p+q)(q+r) : (q+r)^2,$$

or as $p_1:2q_1:r_1$, say.

The interesting question is – in what circumstances will this distribution be the same as that in the generation before? It is easy to see that the condition for this is $q^2 = pr$. And since $q^2 = p_1r_1$, whatever the values of p , q , and r may be, the distribution will in any case continue unchanged after the second generation.

Suppose, to take a definite instance, that A is brachydactyly, and that we start from a population of pure brachydactylous and pure normal persons, say in the ratio of 1:10,000. Then $p = 1$, $q = 0$, $r = 10,000$ and $p_1 = 1$, $q_1 = 10,000$, $r_1 = 100,000,000$. If brachydactyly is dominant, the proportion of brachydactylous persons in the second generation is $20,001:100,020,001$, or practically 2:10,000, twice that in the first generation; and this proportion will afterwards have no tendency whatever to increase. If, on the other hand, brachydactyly were recessive, the proportion in the second generation would be 1:100,020,001, or

practically 1:100,000,000, and this proportion would afterwards have no tendency to decrease.

In a word, there is not the slightest foundation for the idea that a dominant character should show a tendency to spread over a whole population, or that a recessive should tend to die out.

I ought perhaps to add a few words on the effect of the small deviations from the theoretical proportions which will, of course, occur in every generation. Such a distribution as $p_1:2q_1:r_1$, which satisfies the condition $q_1^2 = p_1r_1$, we may call a *stable* distribution. In actual fact we shall obtain in the second generation not $p_1:2q_1:r_1$ but a slightly different distribution $p_2:2q_2:r_2$, which is not "stable." This should, according to theory, give us in the third generation a "stable" distribution $p_3:2q_3:r_3$, also differing from $p_1:2q_1:r_1$; and so on. The sense in which the distribution $p_1:2q_1:r_1$ is "stable" is this, that if we allow for the effects of casual deviations in any subsequent generation, we should, according to theory, obtain at the next generation a new "stable" distribution differing but slightly from the original distribution.

I have, of course, considered only the very simplest hypotheses possible. Hypotheses other than [sic] that of purely random mating will give different results, and, of course, if, as appears to be the case sometimes, the character is not independent of that of sex, or has an influence on fertility, the whole question may be greatly complicated. But such complications seem to be irrelevant to the simple issue raised by Mr. Yule's remarks.

G. H. Hardy
Trinity College, Cambridge,
April 5, 1908

P. S. I understand from Mr. Punnett that he has submitted the substance of what I have said above to Mr. Yule, and that the latter would accept it as a satisfactory answer to the difficulty that he raised. The "stability" of the particular ratio 1:2:1 is recognized by Professor Karl Pearson (*Phil. Trans. Roy. Soc. (A)*, vol. 203, p. 60).

Hardy, G. H. 1908. Mendelian proportions in a mixed population, *Science*, N. S. Vol. XXVIII: 49-50. (letter to the editor)

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... Suppose that Aa is a pair of Mendelian characters, A being dominant, and that in any given generation the numbers of pure dominants (AA), heterozygotes (Aa), and pure recessives (aa) are as $p : 2q : r$. Finally, suppose that **the numbers are fairly large, so that the mating may be regarded as random, that the sexes are evenly distributed among the three varieties, and that all are equally fertile.** A little mathematics of the multiplication-table type is enough to show that in the next generation the numbers will be as

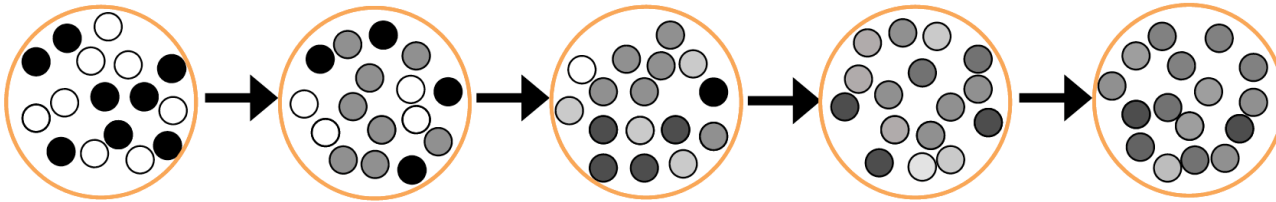
$$(p+q)^2 : 2(p+q)(q+r) : (q+r)^2, \text{ or as } p_1 : 2q_1 : r_1, \text{ say.}$$

whatever the values of p , q , and r may be, the distribution will in any case continue unchanged after the second generation.

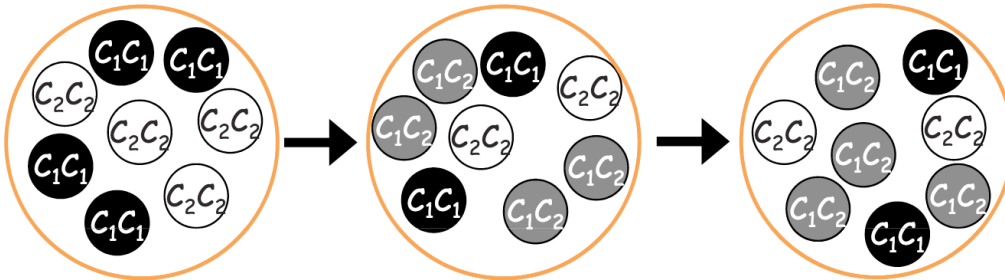
Hardy, G. H. 2003. "Mendelian Proportions in a Mixed Population. 1908." *The Yale Journal of Biology and Medicine* 76 (2): 79-80.

HISTORY OF HARDY-WEINBERG LAW

A) Blending inheritance



B) Mendelian inheritance



1: Masel, Joanna. 2012. "Rethinking Hardy-Weinberg and Genetic Drift in Undergraduate Biology." *BioEssays* 34 (8): 701-10. <https://doi.org/10.1002/bies.201100178>.

HARDY-WEINBERG LAW

Suppose allele frequency of $A = p$, and frequency of $a = q$

$$\begin{array}{ccc} f_{A/A} & f_{A/a} & f_{a/a} \\ \hline p^2 & 2pq & q^2 \end{array}$$

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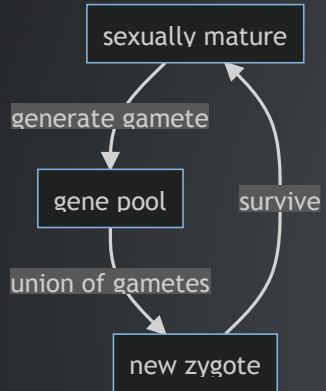
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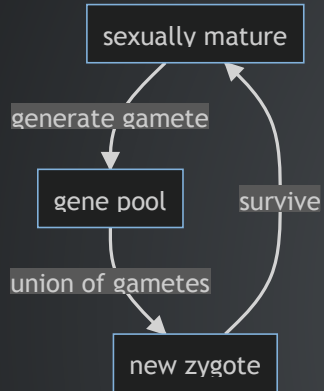
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- Assume the allele frequency of A is the same in females and males.
- The probability of a baby frog to have A/A genotype is p^2
- What other assumptions are needed?
- Will any evolutionary force, such as migration and natural selection, affect the above expectation?

HARDY-WEINBERG LAW

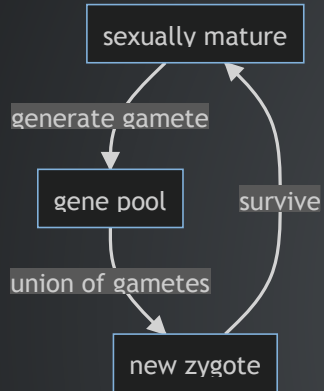


HARDY-WEINBERG LAW



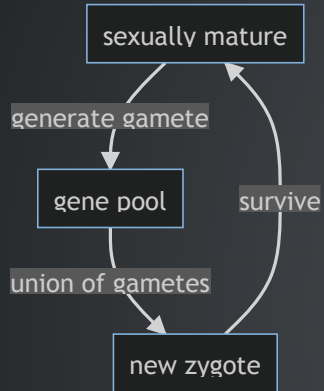
- The prediction of genotype frequency from the allele frequency only relies on the step of forming new zygotes.

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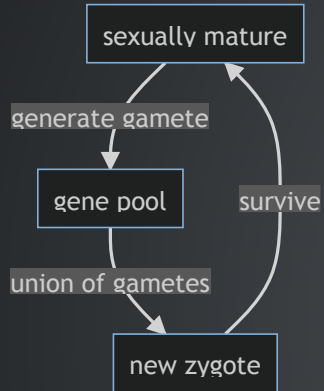
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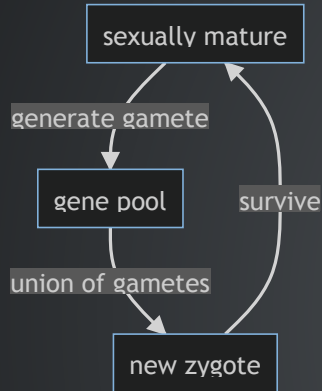
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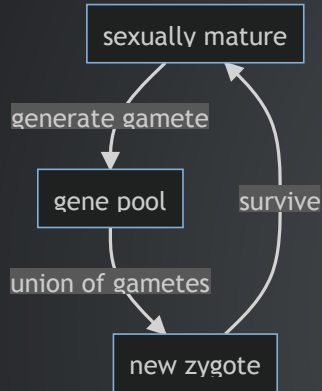
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 - same allele frequencies in both sexes
- Under these assumptions, the expected genotypic frequency distribution is achieved *instantaneously* upon mating
- Often hear about "no selection, no migration" only needed for HW frequencies to represent a **long term equilibrium**, known as Hardy Weinberg Equilibrium

CONSEQUENCES OF HARDY WEINBERG LAW

1.

2.

3.

-

-

-

CONSEQUENCES OF HARDY WEINBERG LAW

1. Genotype frequencies can be estimated easily from allele frequencies.

2.

3.

-

-

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 - This is the Hardy-Weinberg Equilibrium

USING HW LAW TO ESTIMATE HETEROZYGOTE FREQUENCY

Example: cystic fibrosis, an autosomal recessive disease, has an incidence of 1 in 2500 in northern Europeans. Estimate the frequency of carriers.

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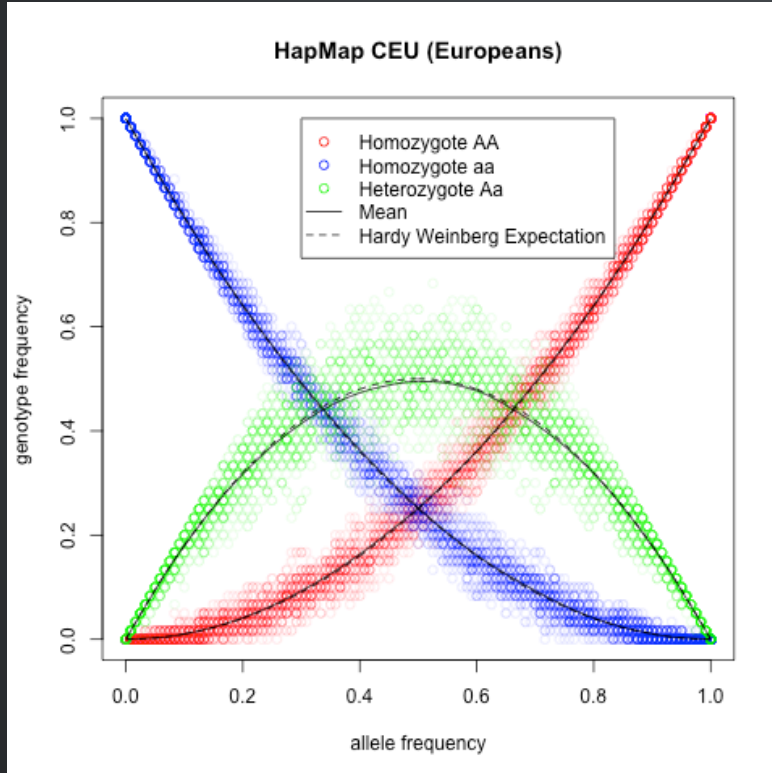
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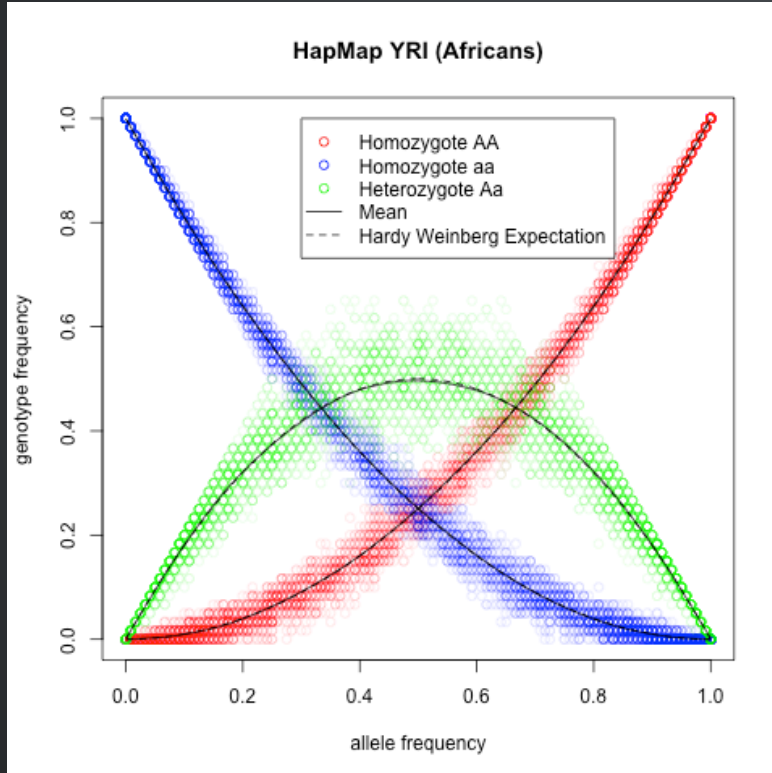
or 1 in 25 people.

HARDY-WEINBERG EQUILIBRIUM



- Solid line: mean genotype frequency; dashed line: HWE prediction. [Source](#)
- While HWE sounds too ideal to be a good fit to real data, researchers found that it actually holds remarkably well within populations. Since the main assumption for HWE to be true is that mating is random, how should one understand this?

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HWE EXERCISE

In a human population, the genotype frequencies at one locus are 0.5 AA, 0.4 Aa, and 0.1 aa. The frequency of the A allele is:

A) 0.20.

B) 0.32.

C) 0.50.

D) 0.70.

E) 0.90.

? Question

On the coastal islands of British Columbia there is a subspecies of black bear (*Ursus americanus kermodei*, Kermode's bear). Many members of this black bear subspecies are white; they're sometimes called spirit bears. These bears aren't hybrids with polar bears, nor are they albinos. They are homozygotes for a recessive change at the MC1R gene. Individuals who are GG at this SNP are white, while AA and AG individuals are black.

The genotype counts for the MC1R polymorphism in a sample of bears from British Columbia's island populations from Ritland et al. (2001) are: $AA = 42$, $AG = 24$, $GG = 21$

What are the expected frequencies of the three genotypes under HWE?

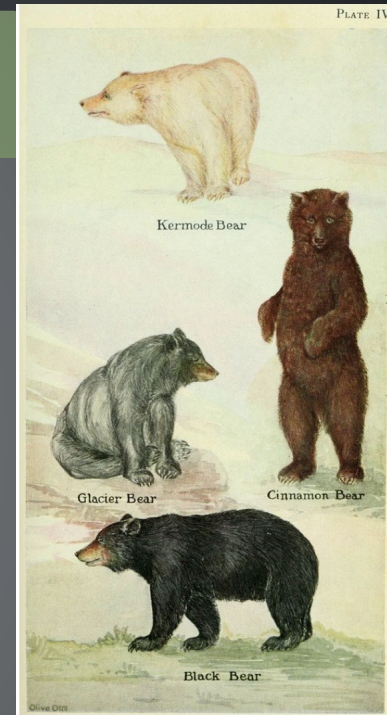


Figure 2.5: Kermode's bear (*Ursus americanus kermodei*). It's possible that this morph is favoured as the salmon these bears eat have a harder time seeing the light morph (KLINKA and REIMCHEN, 2009). The adaptive value of tasting like cinnamon is unknown.

HARDY-WEINBERG LAW FOR X-LINKED LOCI

- How does HW Law apply to X-linked loci?

-
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-
-
-

Male pattern baldness

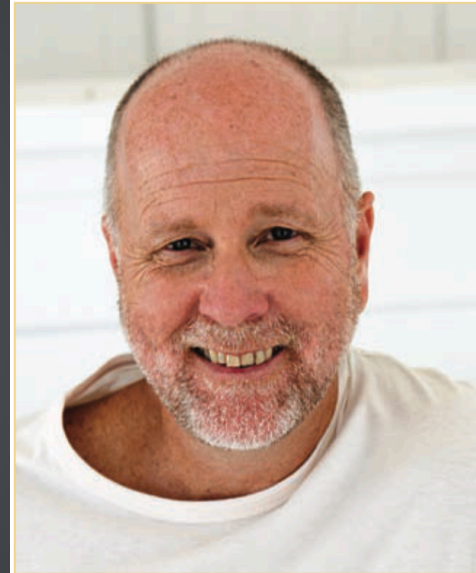


FIGURE 18-9 Individual showing male pattern baldness, an X-chromosome-linked condition. [B2M Productions/Getty Images.]

1: Introduction to Genetic Analysis, ed 11, Fig. 18-9

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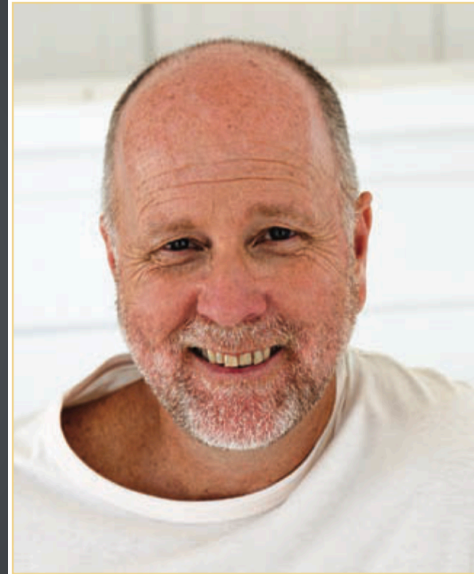


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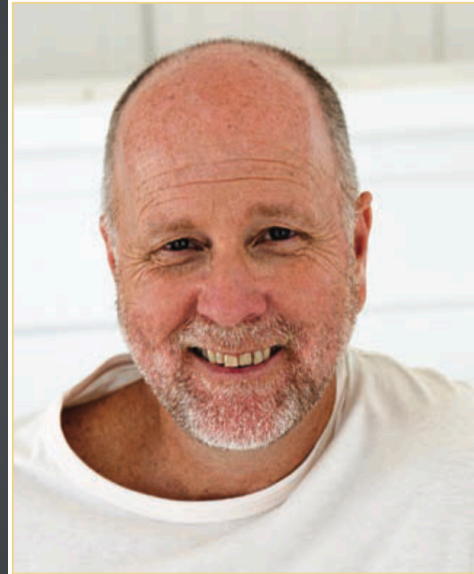


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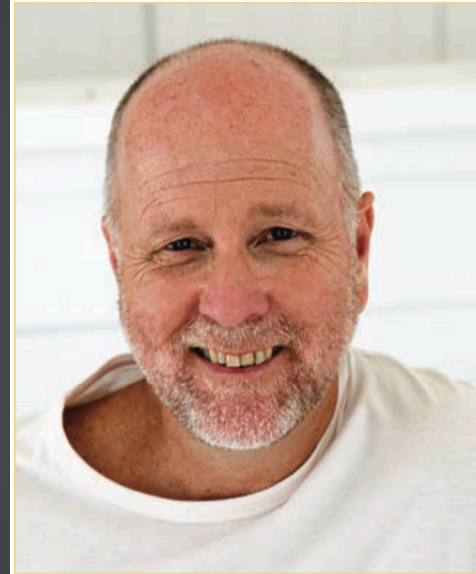


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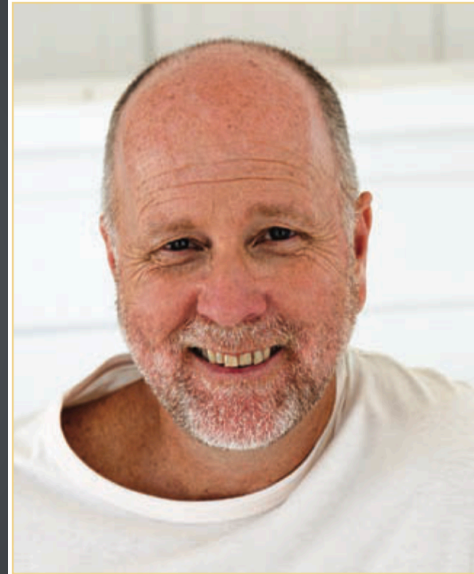


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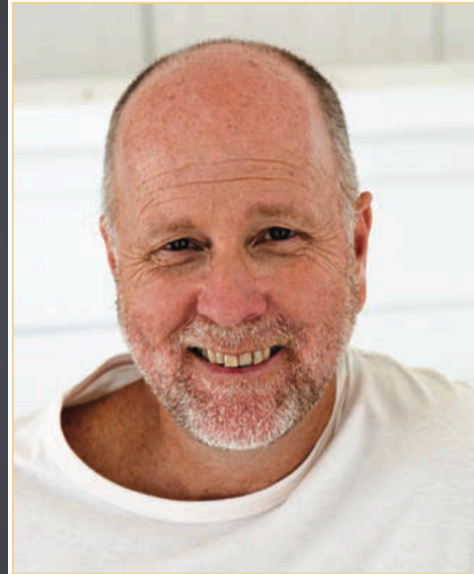


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MAJOR VIOLATIONS TO HWE

ASSORTATIVE MATING

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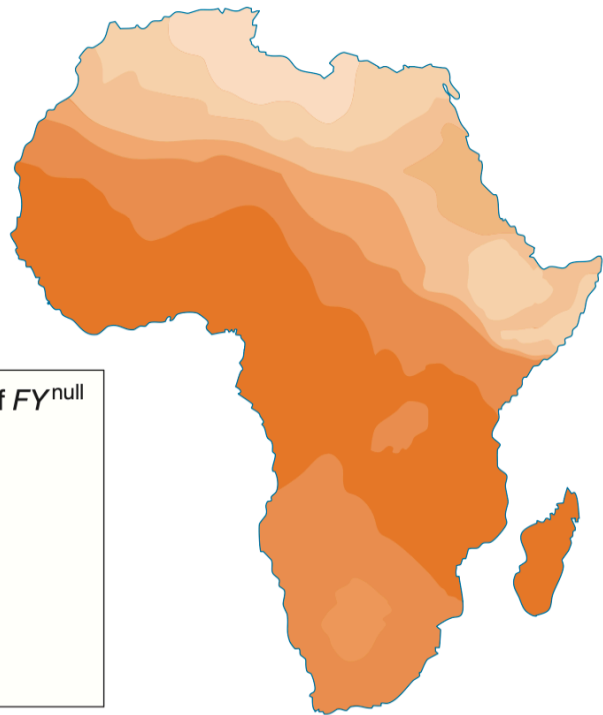
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 - "sweaty shirt test": MHC alleles affect mating choice, prefer individuals with different MHC alleles

MAJOR VIOLATIONS TO HWE

ISOLATION BY DISTANCE

(b)



Frequency variation for the FY^{null} allele of the Duffy blood group locus in Africa. [Data from P. C. Sabeti et al., Science 312, 2006, 1614–1620.]

question

what's the consequence of isolation by distance?

MAJOR VIOLATIONS TO HWE

INBREEDING

MAJOR VIOLATIONS TO HWE

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- Mating between *close* relatives.

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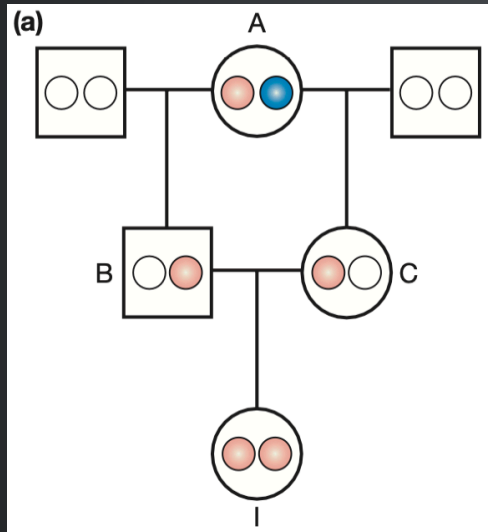
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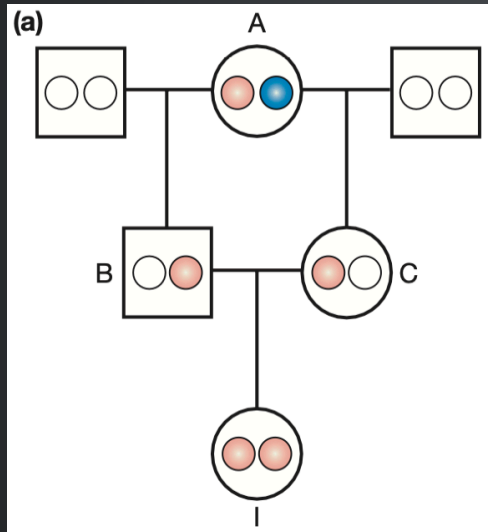
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MEASURING INBREEDING

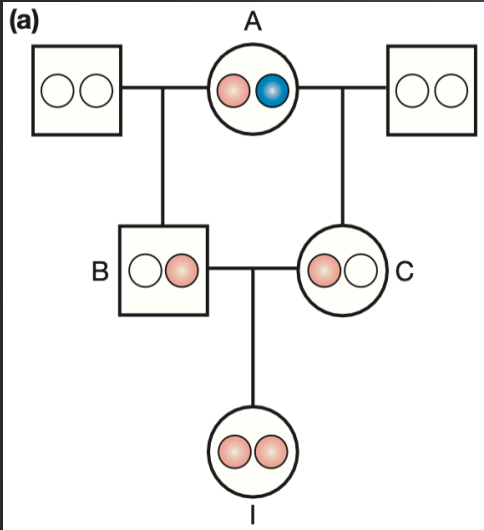


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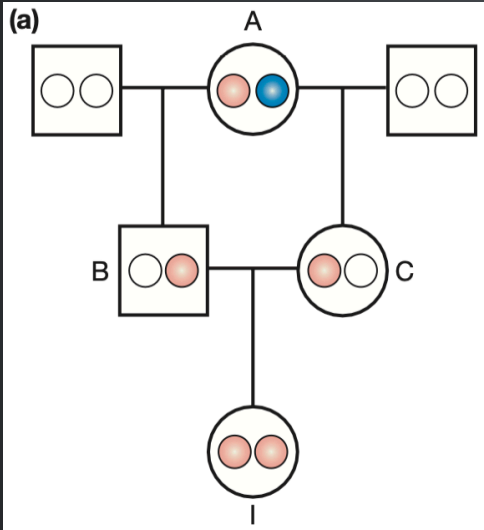
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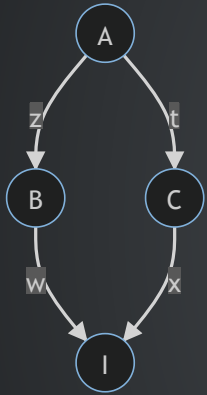
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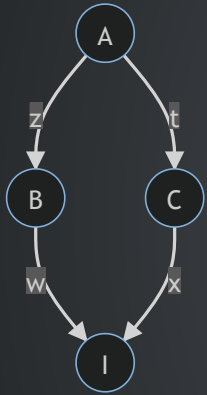
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- A "closed loop" indicates inbreeding.

MEASURING INBREEDING



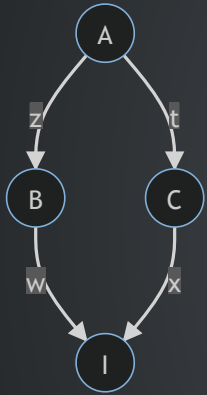
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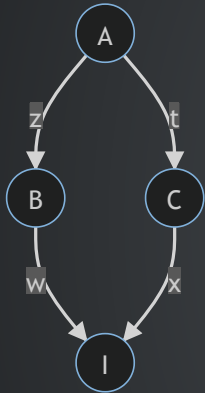
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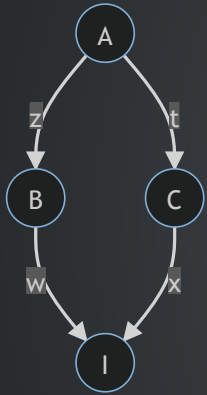
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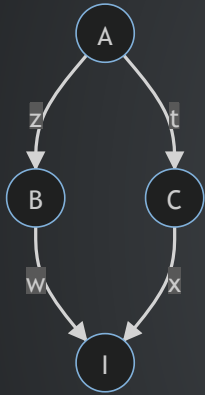
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- $P(x \sim t) = \frac{1}{2}$; $P(w \sim z) = \frac{1}{2}$

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- $P(z \sim t) = \frac{1}{2} + \frac{1}{2}F_A$
- $F_I = P(z \sim t) \times P(w \sim z) \times P(x \sim t) = (\frac{1}{2})^3(1 + F_A)$

BOX 18-2 Calculating Inbreeding Coefficients from Pedigrees

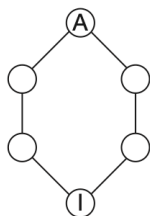
In the main text, we saw that the inbreeding coefficient (F_I) for the offspring of a mating between half-sibs is

$$F_I = \left(\frac{1}{2}\right)^3 (1 + F_A)$$

where F_A is the inbreeding coefficient of the ancestor. This expression includes the term $1/2$ to the third power, $(\frac{1}{2})^3$. In Figure 18-12, you'll see there are three individuals in the inbreeding loop, not counting I. The general formula for computing inbreeding coefficients from pedigrees is

$$F_I = \left(\frac{1}{2}\right)^n (1 + F_A)$$

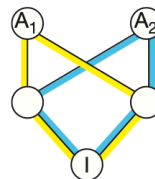
where n is the number of individuals in the inbreeding loop not counting I. Let's look at another pedigree, one in which the grandparents of I are half-sibs:



There are five individuals in the inbreeding loop other than I, so if we assume that the ancestor was not inbred ($F_A = 0$), then

$$F_I = \left(\frac{1}{2}\right)^5 (1 + F_A) = 0.03125$$

In some pedigrees, there is more than one inbreeding loop. Here's a pedigree in which I is the offspring of a mating between full sibs:



For pedigrees with multiple inbreeding loops, you sum the contribution over all of the loops where F_A is the inbreeding coefficient of the ancestor (A) of the given loop:

$$F_I = \sum_{\text{loops}} \left(\frac{1}{2}\right)^n (1 + F_A)$$

Thus, for the pedigree where I is the offspring of a mating between full sibs, we get

$$F_I = \left(\frac{1}{2}\right)^3 (1 + F_{A_1}) + \left(\frac{1}{2}\right)^3 (1 + F_{A_2}) = \frac{1}{4}$$

assuming that the inbreeding coefficients for both ancestors are 0.

CONSEQUENCES OF INBREEDING

Examples of inbreeding effect on frequency of homozygous recessives

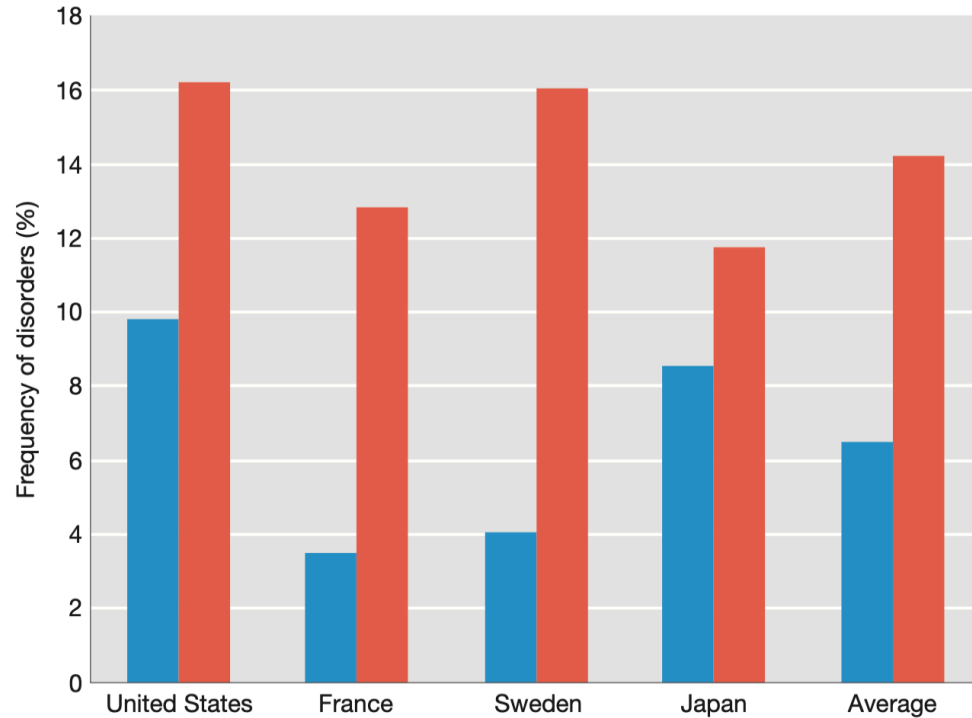
TABLE 18-3 Number of Homozygous Recessives per 10,000 Individuals for Different Allele Frequencies (q)

Mating	F	$q = 0.01$	$q = 0.005$	$q = 0.001$
Unrelated parents	0.0	1.00	0.25	0.01
Parent-offspring or brother-sister	1/4	25.75	12.69	2.51
Half-sib	1/8	13.38	6.47	1.26
First cousin	1/16	7.19	3.36	0.63
Second cousin	1/64	2.55	1.03	0.17

1: Introduction to Genetic Analysis, ed 11, Table 18-3

CONSEQUENCE OF INBREEDING

Inbreeding leads to an increase in recessive genetic disorders



Frequency of genetic disorders among children of unrelated parents (blue columns) compared to that of children of parents who are first cousins (red columns).

Data from C. Stern, Principles of Human Genetics, W. H. Freeman, 1973.

REVIEW OF MONDAY LECTURE

Summary

1. Incorporate the inbreeding coefficient in the calculation of expected genotypic frequency.
2. HWP applied to forensics

“

Microsatellites are regions of the genome where individuals vary for the number of copies of some short DNA repeat that they carry. These regions are often highly variable across individuals, making them a suitable way to identify individuals from a DNA sample. This so-called DNA fingerprinting has a range of applications from establishing paternity and identifying human remains to matching individuals to DNA samples from a crime scene. The FBI make use of the CODIS¹ database. The CODIS database contains the genotypes of over 13 million people, most of whom have been convicted of a crime. Most of the profiles record genotypes at 13 microsatellite loci that are tetranucleotide repeats (since 2017, 20 sites have been genotyped).

¹: CODIS: Combined DNA Index System

The allele counts for two loci (D16S539 and TH01) are shown in the two tables below for a sample of 155 people of European ancestry. You can assume these two loci are on different chromosomes.

You extract a DNA sample from a crime scene. The genotype is 100/80 at the D16S539 locus and 70/93 at TH01.

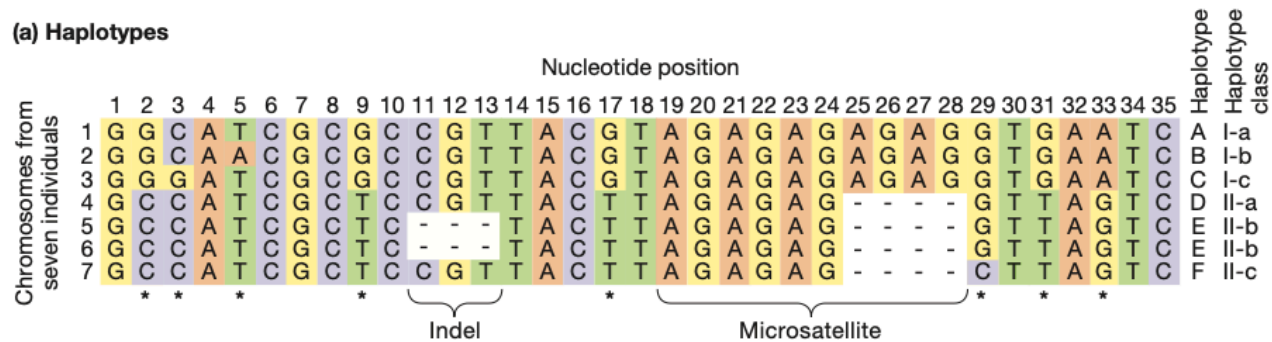
A) You have a suspect in custody. Assuming this suspect is innocent and of European ancestry, what is the probability that their genotype would match this profile by chance (a false-match probability)?

B) The FBI uses ≥ 13 markers. Why is this higher number necessary to make the match statement convincing evidence in court?

C) An early case that triggered debate among forensic geneticists was a crime among the Abenaki, a Native American community in Vermont (see Lewontin, 1994, for discussion). There was a DNA sample from the crime scene, and the perpetrator was thought likely to be a member of the Abenaki community. Given that allele frequencies vary among populations, why would people be concerned about using data from a non-Abenaki population to compute a false match probability?

HOW DO YOU QUANTIFY THE LEVEL OF GENETIC VARIATION?

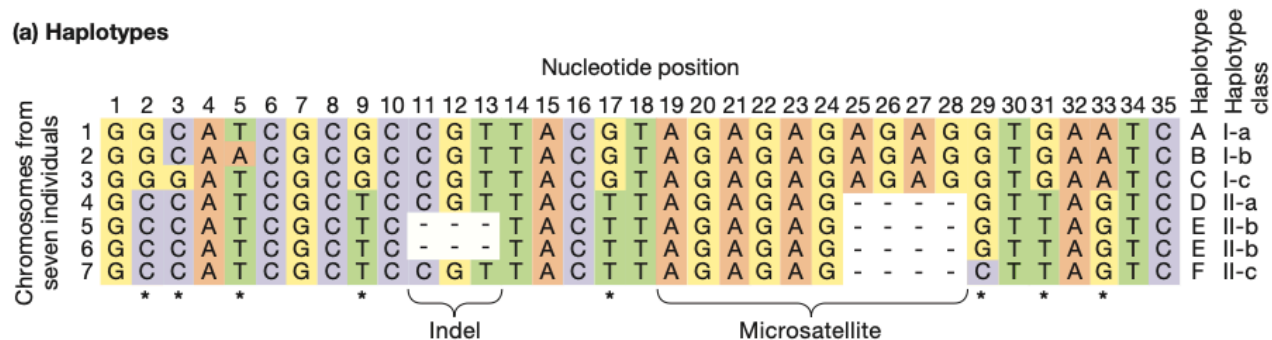
A haplotype network shows the relationship among haplotypes



- count the number of variable sites, or k

HOW DO YOU QUANTIFY THE LEVEL OF GENETIC VARIATION?

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(a) Haplotypes

(a) Haplotypes																																			Haplotype	Haplotype class		
Chromosomes from seven individuals	Nucleotide position																																					
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35			
	1	G	G	C	A	T	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	A	I-a
	2	G	G	C	A	A	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	B	I-b
	3	G	G	G	A	T	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	C	I-c
	4	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	D	II-a
	5	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	E	II-b
	6	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	F	II-b
	7	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	C	T	T	A	G	T	C	F	II-c
		*	*		*			*		Indel						*	Microsatellite											*		*								

Indel

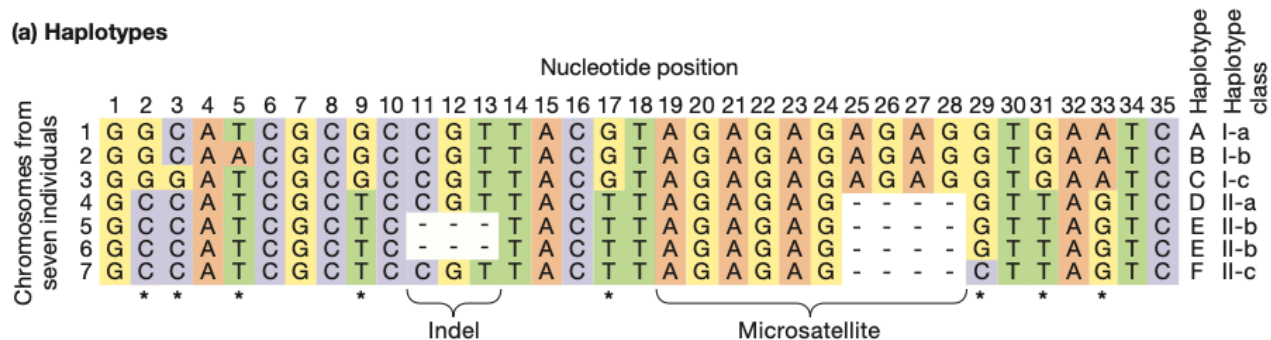
Microsatellite

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- account for the number of sequences (sample size)

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A haplotype network shows the relationship among haplotypes

(a) Haplotypes



- count the number of variable sites, or k
- account for the length of the region (sequence)
- account for the number of sequences (sample size)
- Average # of differences between two sequences

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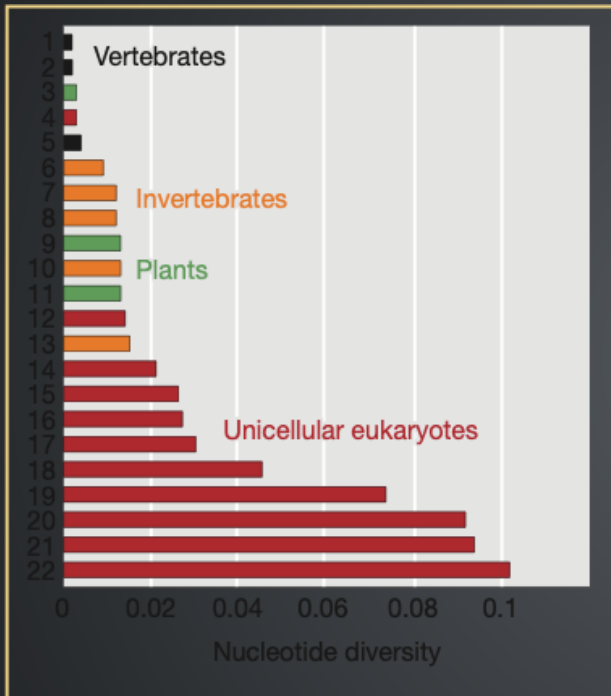
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- **Nucleotide Diversity**

- probability that any nucleotide position is different between two randomly selected allele from a population.
- like gene diversity, but calculated per nucleotide position and averaged
- independent of length and more easily compared across species

LEVEL OF GENETIC VARIATION VARY ACROSS SPECIES



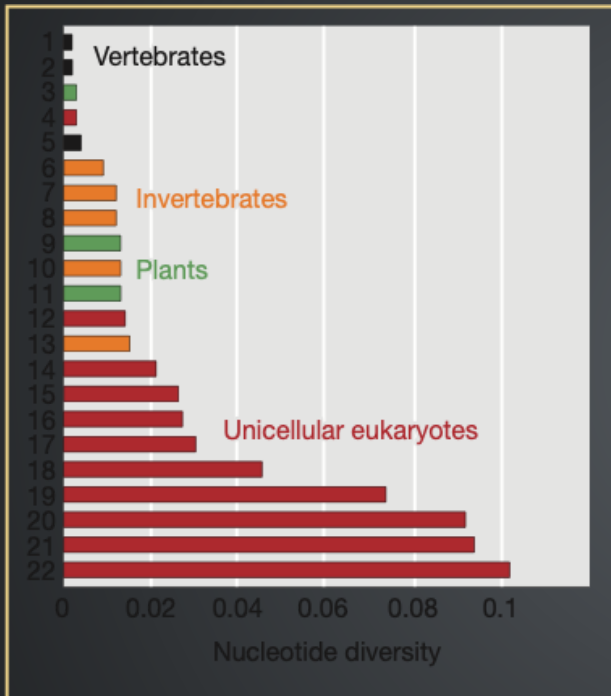
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$$\theta = 4N_e\mu$$

-
-

M. Lynch and J. S. Conery, Science 2003

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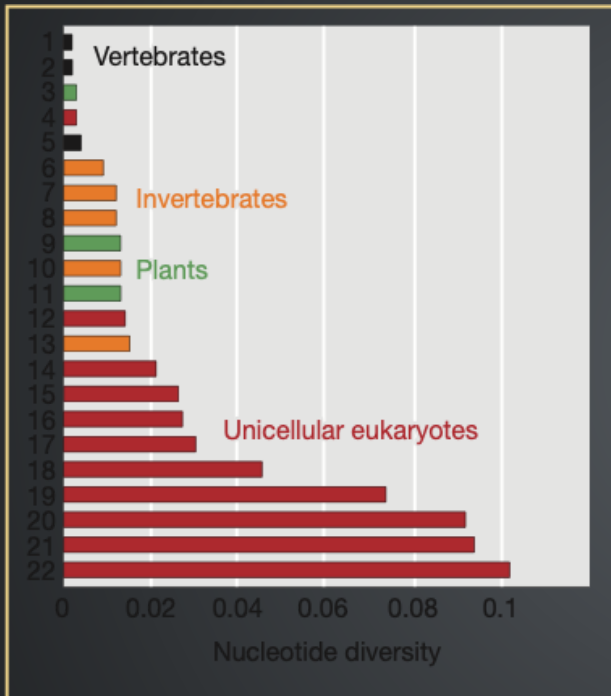
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- neutral theory predicts the level of genetic diversity scales with the effective population size

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- the range of genetic diversity in a sample of 167 species is 0.01%-8%
- this is likely several orders of magnitude smaller than the range of effective population size among species (think some vertebrate species with fewer than 100 breeding individuals vs plants or bacteria with a very large census population size)

M. Lynch and J. S. Conery, Science 2003

WHAT FORCES DRIVE BIOLOGICAL EVOLUTION?

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- How do new alleles enter the gene pool?
- How do existing variation give rise to new combinations?
- What forces drive changes in the frequency of alleles?
- We will examine the following five forces in detail
 1. mutation
 2. migration
 3. random genetic drift
 4. recombination
 5. selection

MUTATION

TABLE 18-5 Approximate Mutation Rates per Generation per Haploid Genome

Organism	SNP mutations (per bp)	Microsatellite
<i>Arabidopsis</i>	7×10^{-9}	9×10^{-4}
Maize	3×10^{-8}	8×10^{-4}
<i>E. coli</i>	5×10^{-10}	—
Yeast	5×10^{-10}	4×10^{-5}
<i>C. elegans</i>	3×10^{-9}	4×10^{-3}
<i>Drosophila</i>	4×10^{-9}	9×10^{-6}
Mouse	4×10^{-9}	3×10^{-4}
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Source: Data from multiple published studies.

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- data in human (circa 2009) gave an estimate of $\sim 3.0 \times 10^{-8}$ mutations/nucleotide/generation for a part of the Y-chromosome. if we extrapolate this to the entire human genome, we get an estimate of about 100 new mutations one would inherit from each of our parents *on average*.

MIGRATION

- In addition to mutation, migration is the other way by which new variation can be introduced into a population (not counting **new combinations**, see recombination)
- when there is isolation by distance, migration (or gene flow) is a homogenizing force, preventing allele frequencies from diverging too far

RECOMBINATION

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-

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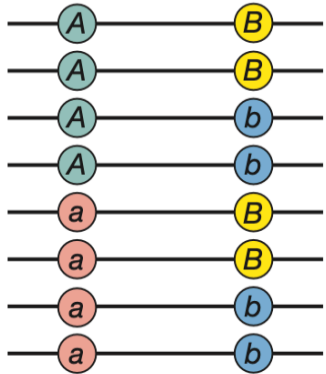
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 - If not, then the loci are said to be in **linkage disequilibrium (LD)**.

LINKAGE DISEQUILIBRIUM

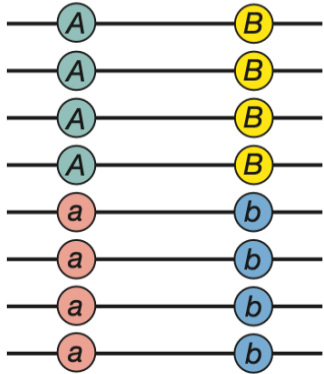
Linkage disequilibrium is the nonrandom association between two loci

(a) Linkage equilibrium



$$\begin{aligned} p_A &= 0.5 & P_{AB} &= 0.25 \\ p_a &= 0.5 & P_{Ab} &= 0.25 \\ p_B &= 0.5 & P_{aB} &= 0.25 \\ p_b &= 0.5 & P_{ab} &= 0.25 \end{aligned}$$

(b) Linkage disequilibrium

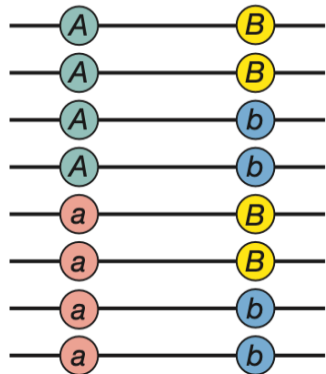


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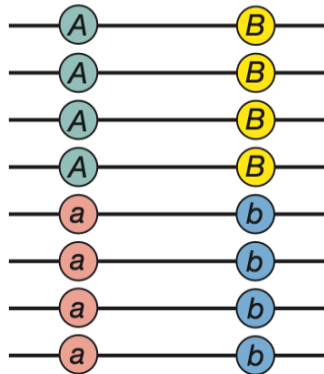
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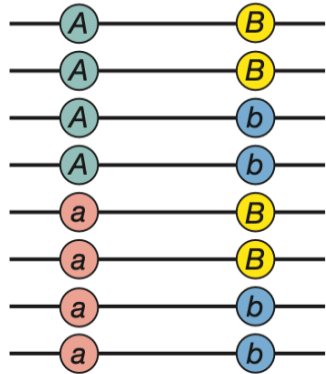
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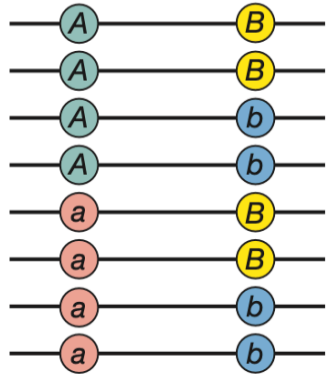
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- Define LD as the difference (D) between the observed and expected frequencies

$$D = P_{AB} - p_A p_B$$

- Exercise:** calculate D for the two scenarios on the left

LINKAGE DISEQUILIBRIUM

- How LD arises

Nucleotide variation at the *G6PD* gene in humans

Individual	Origin	Allele	SNP																		Haplotype
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
			A	G	A	C	C	G	C	C	C	C	C	G	G	C	T	C	A	C	
1	Southern African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
2	Central African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
3	Central African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
4	African American	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
5	African American	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
6	Central African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
7	Central African	A+	G	.	G	.	.	.	.	.	T	.	.	.	.	C	.	G	.	2	
8	Central African	A+	G	.	G	.	.	.	.	.	T	.	.	.	.	C	.	G	.	2	
9	Central African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	C	.	G	.	3	
10	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	A	.	.	C	T	G	4	
11	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	A	.	.	C	T	G	4	
12	Southern African	B	.	.	.	T	.	.	.	.	.	.	.	.	.	.	C	T	G	5	
13	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	C	T	G	6	
14	Southern African	B	.	.	.	.	.	.	.	.	.	T	.	A	.	.	C	.	G	7	
15	Central African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	8	
16	European	B	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	8	
17	European	B	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	8	
18	European	B	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	8	
19	Southwest Asian	B	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	8	
20	East Asian	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	C	.	G	3	
21	Native American	B	.	.	.	.	.	A	T	.	.	.	.	.	.	.	C	.	G	9	
22	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	
23	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	
24	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	
25	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	

LINKAGE DISEQUILIBRIUM

Nucleotide variation at the *G6PD* gene in humans

Individual	Origin	Allele	SNP																		Haplotype	
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		
			A	G	A	C	C	G	C	C	C	C	C	G	G	C	T	C	A	C		
1	Southern African	A-	G	A	G	.	G	.	.	T	.	T	.	.	.	.	C	.	G	.	1	
2	Central African	A-	G	A	G	.	G	.	.	T	.	T	.	.	.	.	C	.	G	.	1	
3	Central African	A-	G	A	G	.	G	.	.	T	.	T	.	.	.	.	C	.	G	.	1	
4	African American	A-	G	A	G	.	G	.	.	T	.	T	.	.	.	.	C	.	G	.	1	
5	African American	A-	G	A	G	.	G	.	.	T	.	T	.	.	.	.	C	.	G	.	1	
6	Central African	A-	G	A	G	.	G	.	.	T	.	T	.	.	.	.	C	.	G	.	1	
7	Central African	A+	G	.	G	.	.	.	.	.	T	.	.	.	.	.	C	.	G	.	2	
8	Central African	A+	G	.	G	.	.	.	.	.	T	.	.	.	.	.	C	.	G	.	2	
9	Central African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	C	.	G	.	3	
10	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	A	.	.	C	T	G	.	4
11	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	A	.	.	C	T	G	.	4
12	Southern African	B	.	.	.	T	.	.	.	.	.	.	.	.	.	.	.	C	T	G	.	5
13	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	C	T	G	.	6
14	Southern African	B	.	.	.	.	.	.	.	.	.	.	T	.	A	.	.	C	.	G	.	7
15	Central African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8
16	European	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8
17	European	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8
18	European	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8
19	Southwest Asian	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8
20	East Asian	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	C	.	G	.	3
21	Native American	B	.	.	.	.	.	.	A	T	.	.	.	.	.	.	.	C	.	G	.	9
22	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10
23	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10
24	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10
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			A	G	A	C	C	G	C	C	C	C	C	G	G	C	T	C	A	C	
1	Southern African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
2	Central African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
3	Central African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
4	African American	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
5	African American	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
6	Central African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
7	Central African	A+	G	.	G	.	.	.	.	.	T	.	.	.	.	C	.	G	.	2	
8	Central African	A+	G	.	G	.	.	.	.	.	T	.	.	.	.	C	.	G	.	2	
9	Central African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	C	.	G	.	3	
10	Southern African	B	.	.	.	.	.	.	.	.	.	.	A	.	.	C	T	G	.	4	
11	Southern African	B	.	.	.	.	.	.	.	.	.	.	A	.	.	C	T	G	.	4	
12	Southern African	B	.	.	.	T	.	.	.	.	.	.	.	.	.	C	T	G	.	5	
13	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	C	T	G	.	6	
14	Southern African	B	.	.	.	.	.	.	.	.	.	T	.	A	.	C	.	G	.	7	
15	Central African	B	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8	
16	European	B	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8	
17	European	B	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8	
18	European	B	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8	
19	Southwest Asian	B	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8	
20	East Asian	B	.	.	.	.	.	.	.	.	.	.	.	.	.	C	.	G	.	3	
21	Native American	B	.	.	.	.	.	A	T	.	.	.	.	.	.	C	.	G	.	9	
22	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	
23	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	
24	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	
25	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	

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			A	G	A	C	C	G	C	C	C	C	C	G	G	C	T	C	A	C	
1	Southern African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
2	Central African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
3	Central African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
4	African American	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
5	African American	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
6	Central African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
7	Central African	A+	G	.	G	.	.	.	.	.	T	.	.	.	.	C	.	G	.	2	
8	Central African	A+	G	.	G	.	.	.	.	.	T	.	.	.	.	C	.	G	.	2	
9	Central African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	C	.	G	.	3	
10	Southern African	B	.	.	.	.	.	.	.	.	.	.	A	.	.	C	T	G	.	4	
11	Southern African	B	.	.	.	.	.	.	.	.	.	.	A	.	.	C	T	G	.	4	
12	Southern African	B	.	.	.	T	.	.	.	.	.	.	.	.	.	C	T	G	.	5	
13	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	C	T	G	.	6	
14	Southern African	B	.	.	.	.	.	.	.	.	.	T	.	A	.	C	.	G	.	7	
15	Central African	B	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8	
16	European	B	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8	
17	European	B	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8	
18	European	B	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8	
19	Southwest Asian	B	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8	
20	East Asian	B	.	.	.	.	.	.	.	.	.	.	.	.	.	C	.	G	.	3	
21	Native American	B	.	.	.	.	.	A	T	.	.	.	.	.	.	C	.	G	.	9	
22	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	
23	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	
24	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	
25	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	

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			A	G	A	C	C	G	C	C	C	C	C	G	G	C	T	C	A	C	
1	Southern African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
2	Central African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
3	Central African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
4	African American	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
5	African American	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
6	Central African	A-	G	A	G	.	G	.	T	.	T	.	.	.	.	C	.	G	.	1	
7	Central African	A+	G	.	G	.	.	.	.	.	T	.	.	.	.	C	.	G	.	2	
8	Central African	A+	G	.	G	.	.	.	.	.	T	.	.	.	.	C	.	G	.	2	
9	Central African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	C	.	G	.	3	
10	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	A	.	.	C	T	G	4	
11	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	A	.	.	C	T	G	4	
12	Southern African	B	.	.	.	T	.	.	.	.	.	.	.	.	.	.	C	T	G	5	
13	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	C	T	G	6	
14	Southern African	B	.	.	.	.	.	.	.	.	.	T	.	A	.	.	C	.	G	7	
15	Central African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	8	
16	European	B	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	8	
17	European	B	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	8	
18	European	B	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	8	
19	Southwest Asian	B	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	8	
20	East Asian	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	C	.	G	3	
21	Native American	B	.	.	.	.	.	.	A	T	.	.	.	.	.	.	C	.	G	9	
22	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	
23	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	
24	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	
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			A	G	A	C	C	G	C	C	C	C	C	G	G	C	T	C	A	C	
1	Southern African	A-	G	A	G	.	G	.	.	T	.	T	.	.	.	.	C	.	G	.	1
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10	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	A	.	.	C	T	G	.	4
11	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	A	.	.	C	T	G	.	4
12	Southern African	B	.	.	.	T	.	.	.	.	.	.	.	.	.	.	C	T	G	.	5
13	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	C	T	G	.	6
14	Southern African	B	.	.	.	.	.	.	.	.	.	T	.	A	.	.	C	.	G	.	7
15	Central African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8
16	European	B	.	.	.	.	.	.	.	.	.	.	.	.	.	T	C	.	G	.	8
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21	Native American	B	.	.	.	.	.	.	A	T	.	.	.	.	.	.	C	.	G	.	9
22	Southern African	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10
23	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10
24	Native American	B	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10
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- can be used to estimate the age of mutation, how?

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 - suppose this species can self-fertilize, and that the population size remains at one in the next generation (t_1). what's the probability that the allele frequency will change ("drift") to something other than its original value?

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 - how about at generation t_2 ?

GENETIC DRIFT AND POPULATION SIZE

- When calculating the expected genotype frequency from allele frequencies under Hardy-Weinberg Equilibrium, we made a sneaky assumption - the population size is assumed to be so large that we can "sample with replacement".
- Real populations have finite population sizes. Sampling can lead to fluctuations in the allele frequency from generation to generation
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 - what happens when we increase the population size (N) to 2?

GENETIC DRIFT AND POPULATION SIZE

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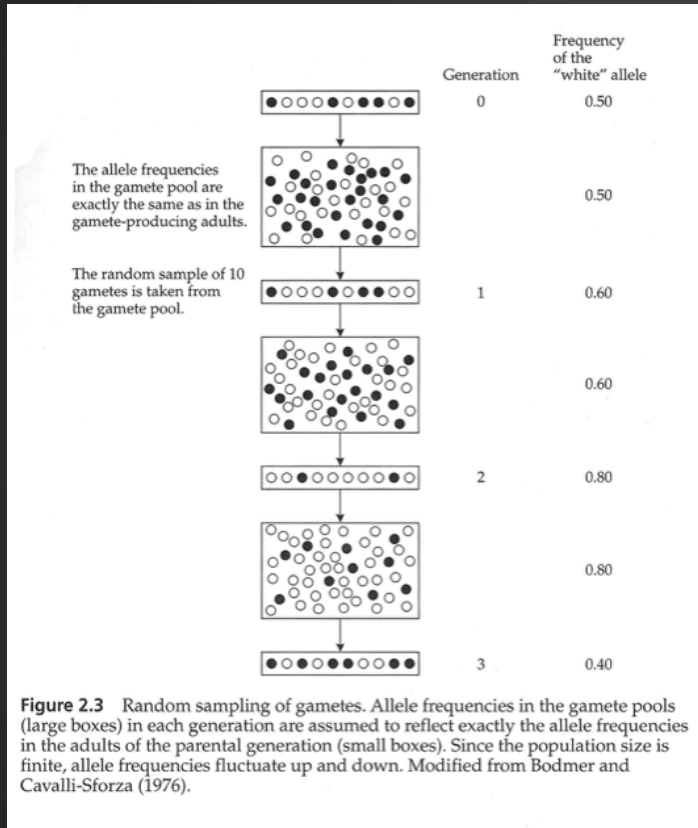
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- $1/2N \ll 1$, $H_t = H_0 e^{-t/2N}$ (can you prove this?)

GENETIC DRIFT AND POPULATION SIZE

Drift

any change in allele frequencies due to sampling error, not just loss or fixation of an allele

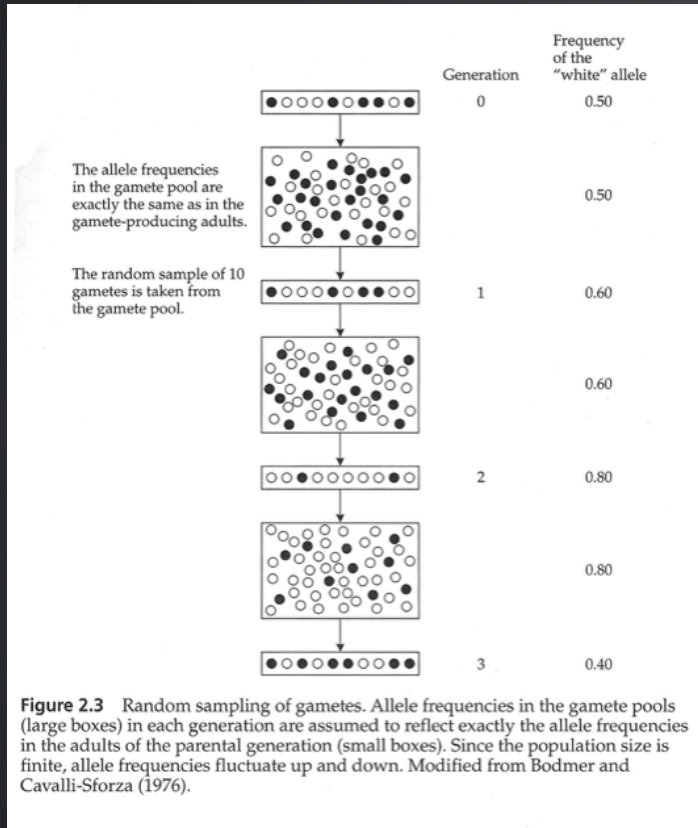


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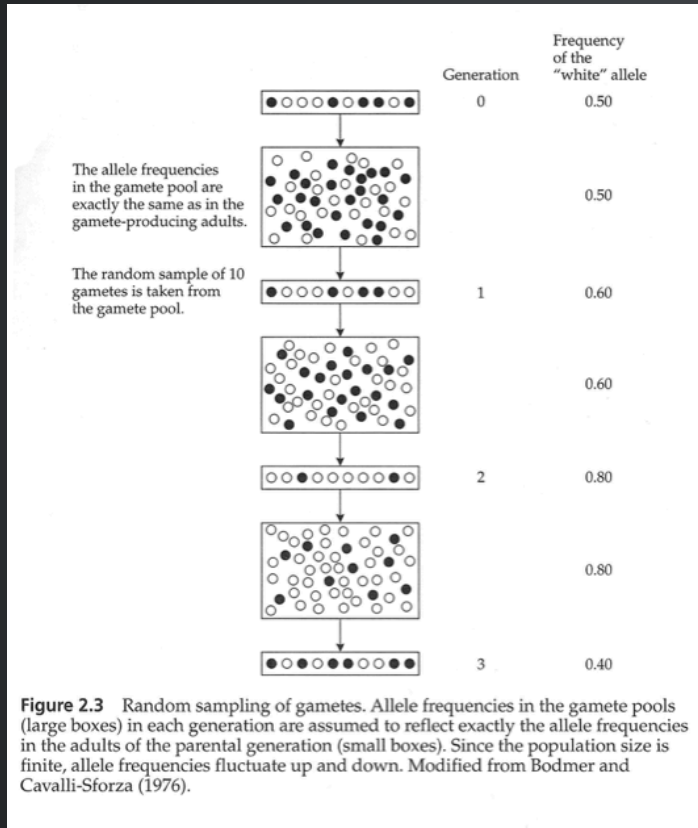


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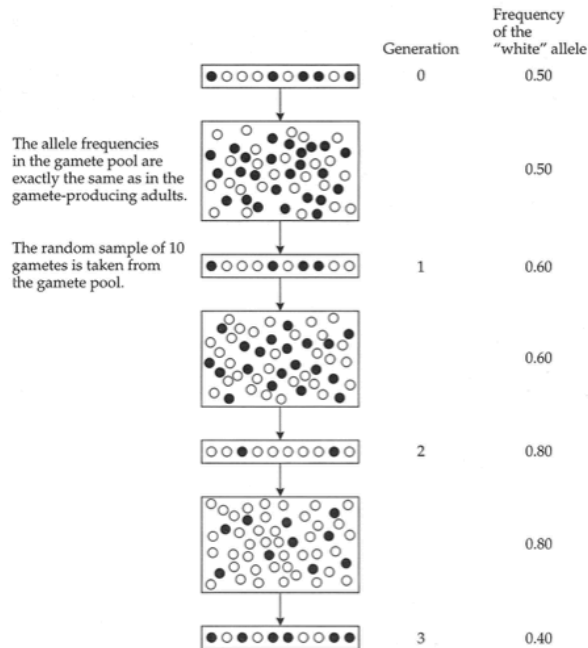
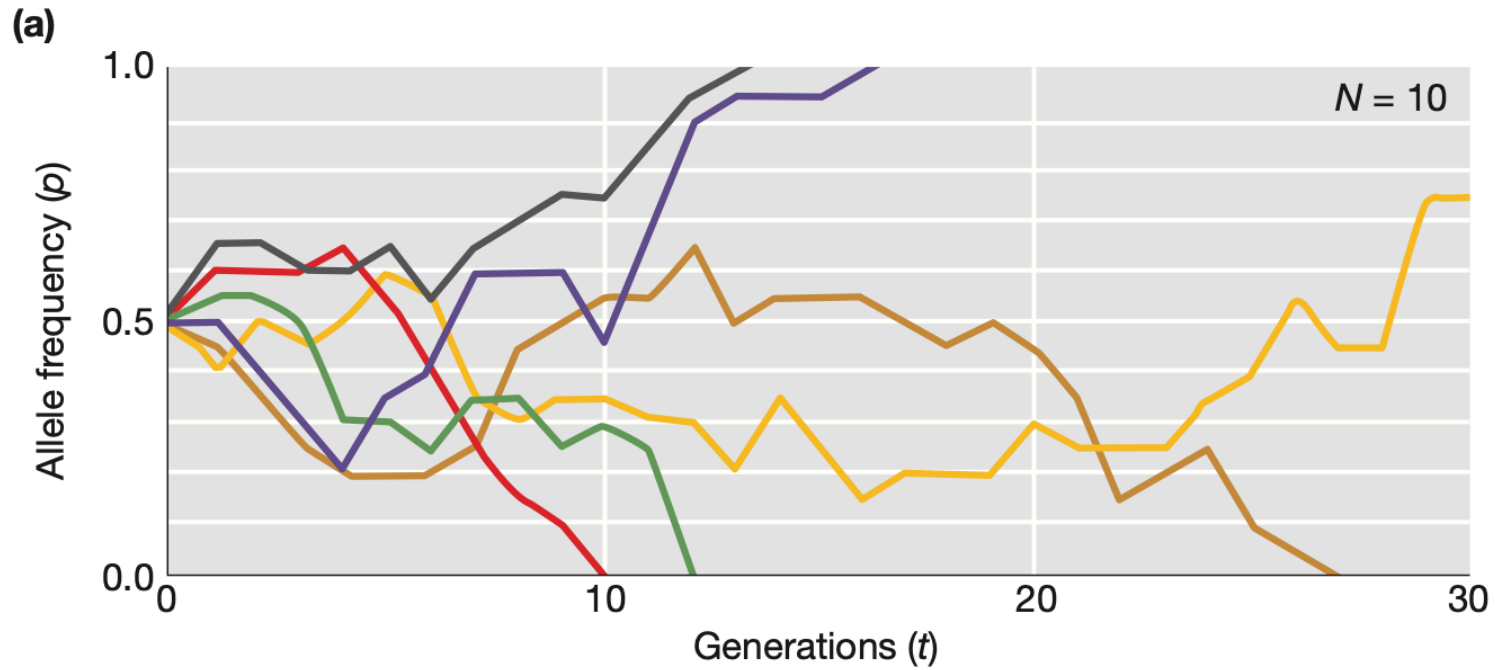
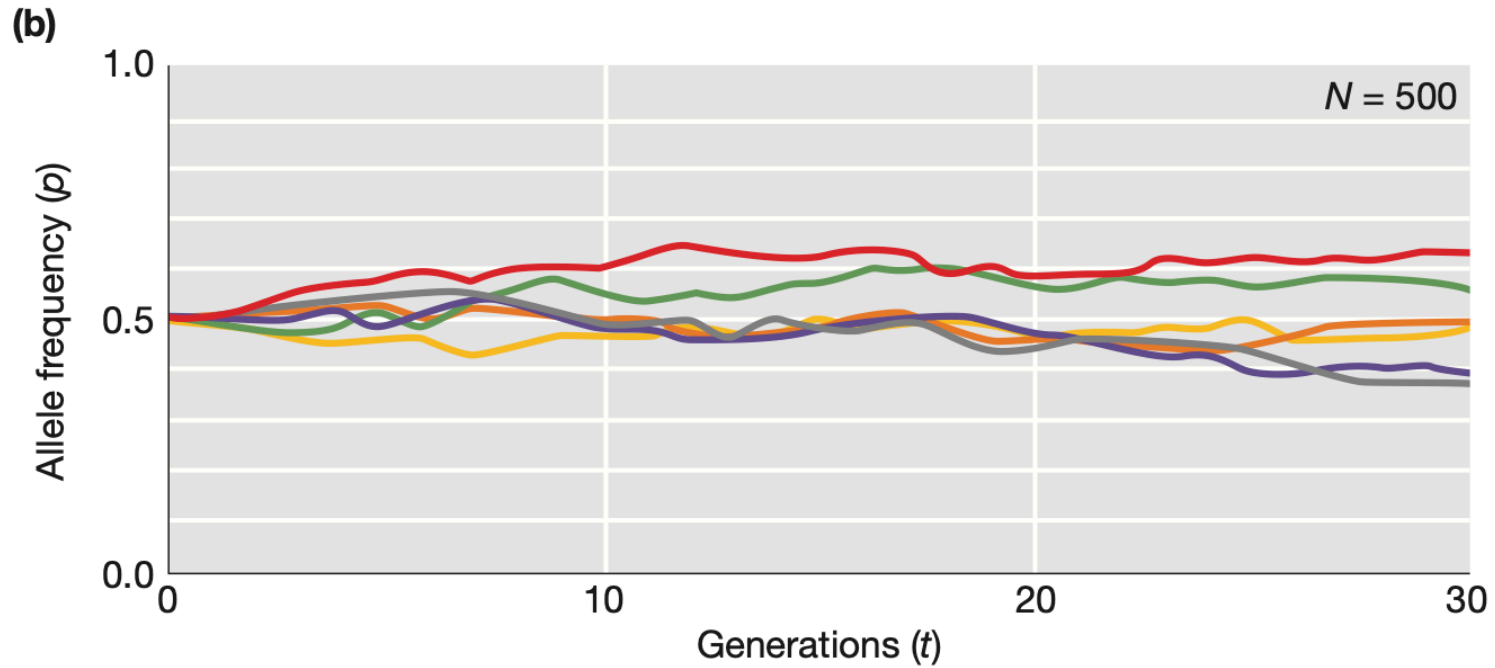


Figure 2.3 Random sampling of gametes. Allele frequencies in the gamete pools (large boxes) in each generation are assumed to reflect exactly the allele frequencies in the adults of the parental generation (small boxes). Since the population size is finite, allele frequencies fluctuate up and down. Modified from Bodmer and Cavalli-Sforza (1976).

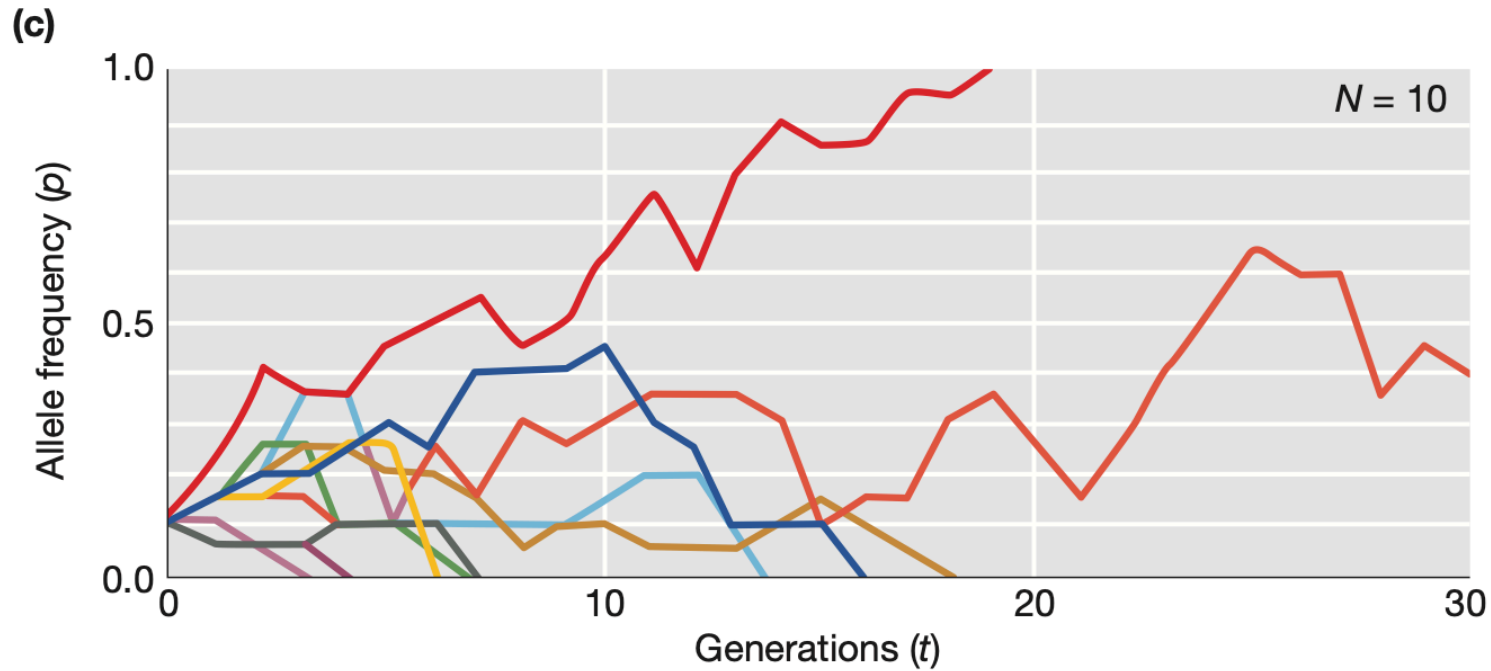
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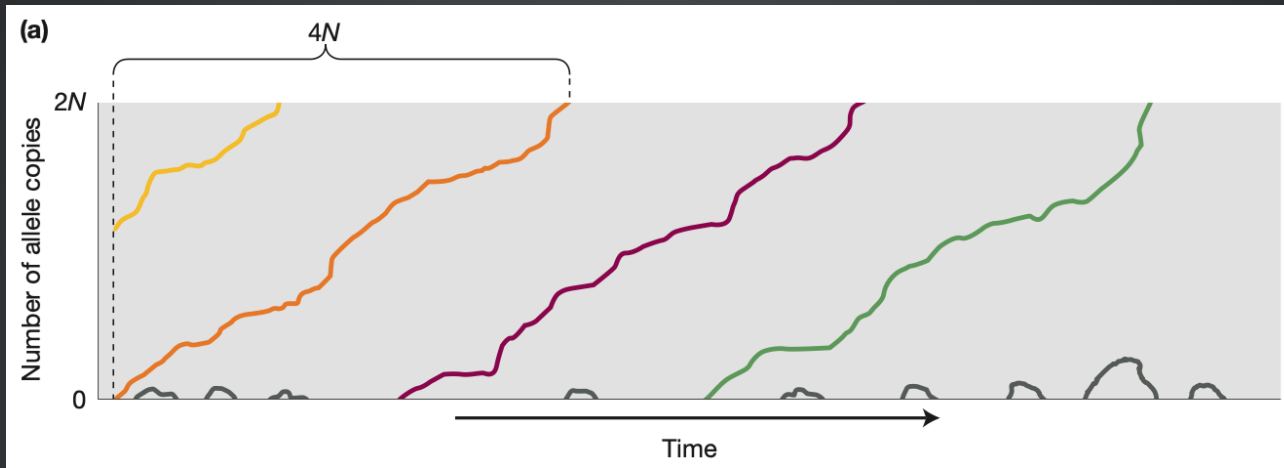
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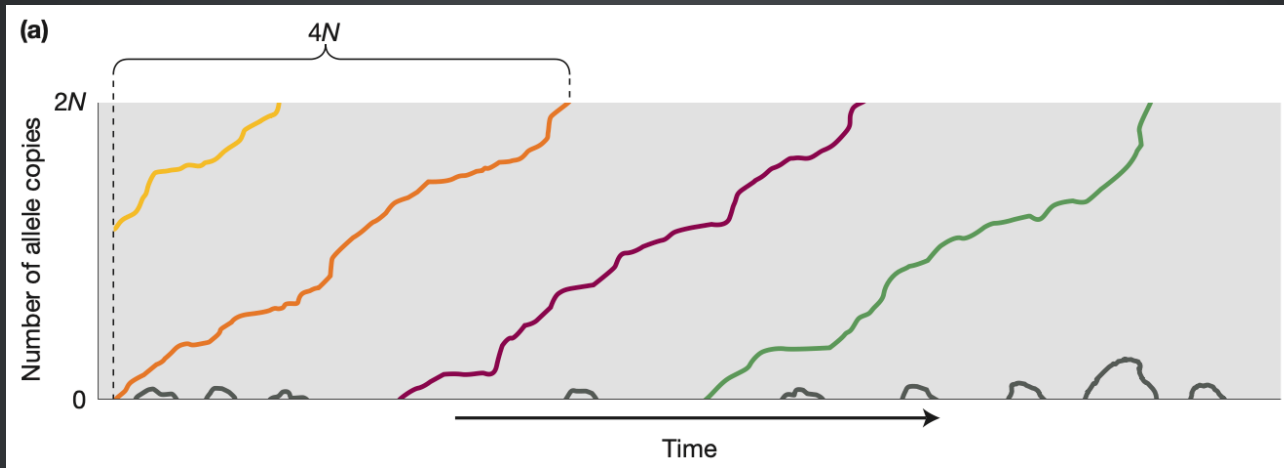
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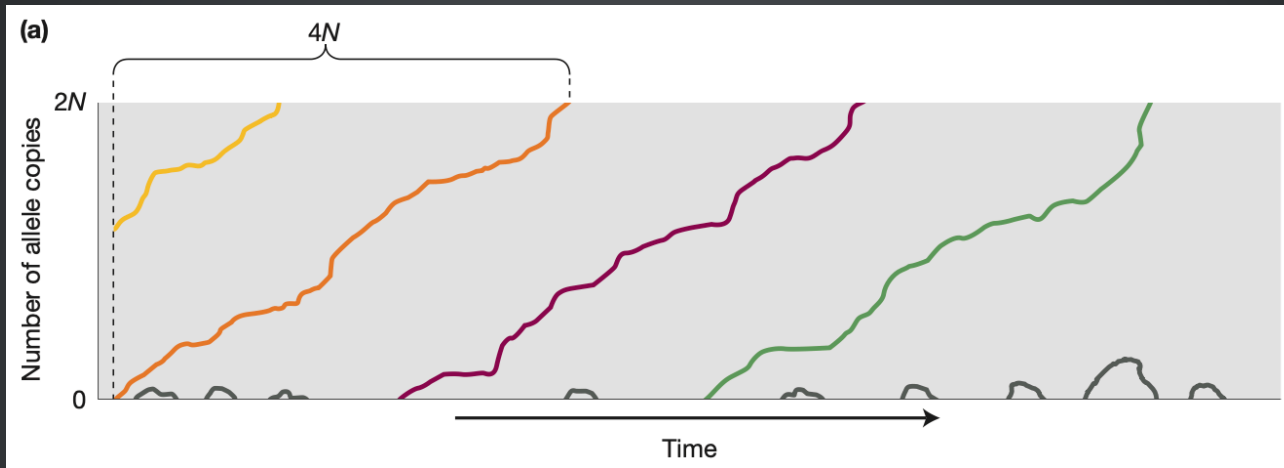


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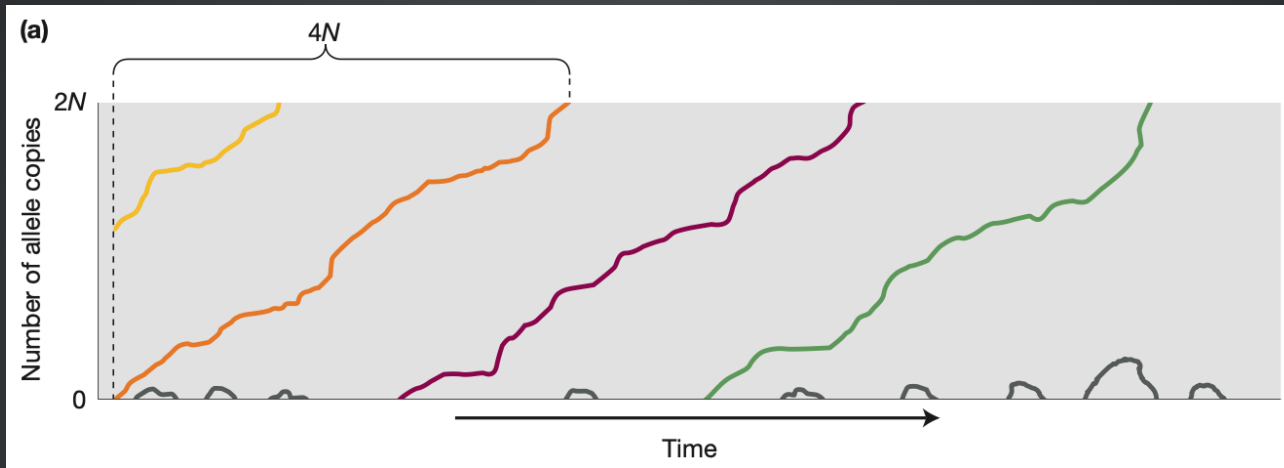
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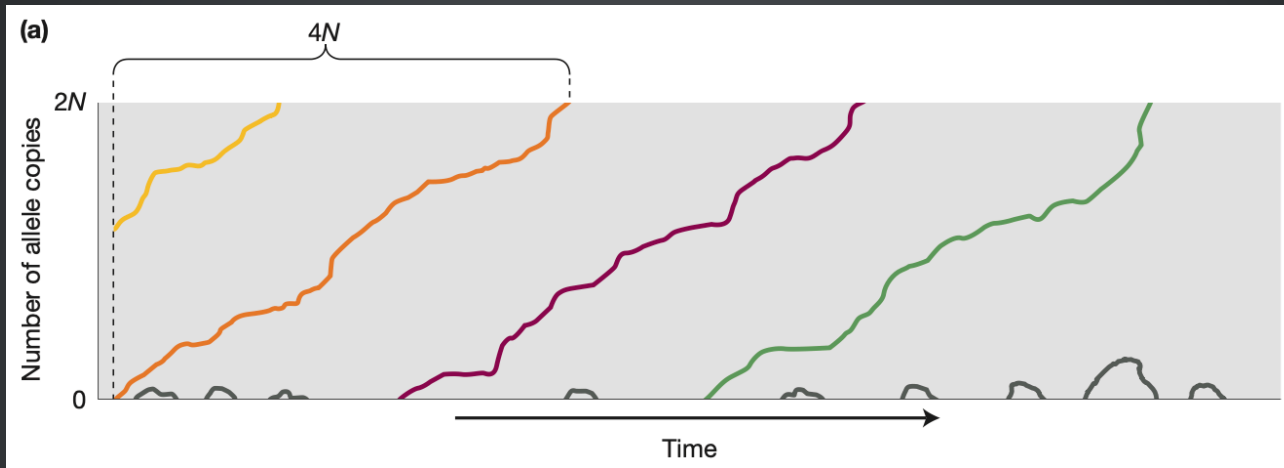
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CONSEQUENCE OF GENETIC DRIFT



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 - or it may not pass on the beneficial mutation (since by definition, a new mutation must be a single copy in a diploid individual)

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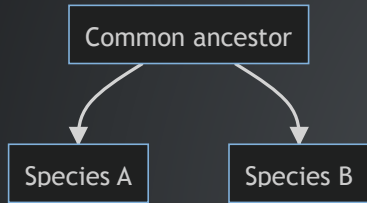
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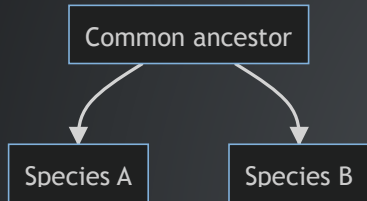
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- that is, the substitution rate for neutral mutations equal the rate at which they appear. this is only true for neutral mutations!

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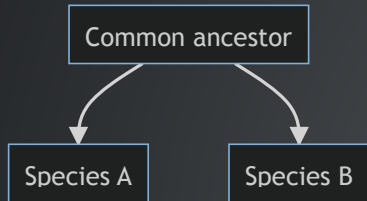
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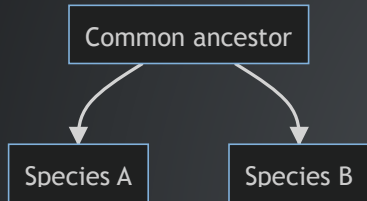
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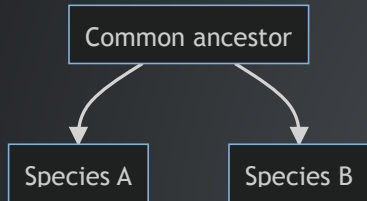
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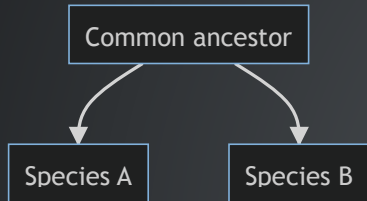
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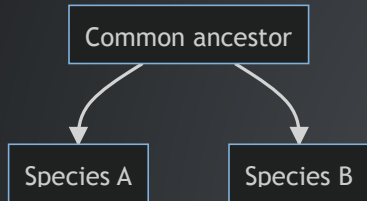
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GENETIC DRIFT AND POPULATION SIZE



Key concept

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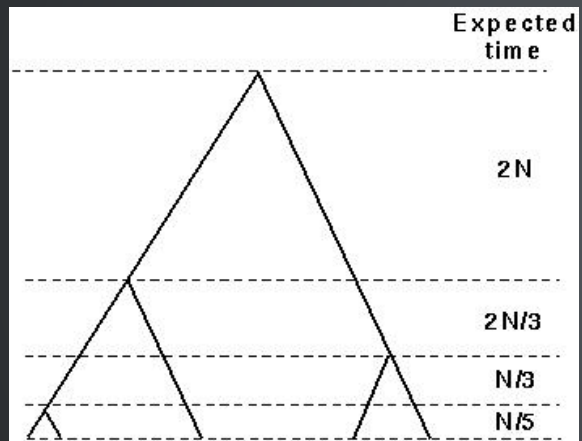
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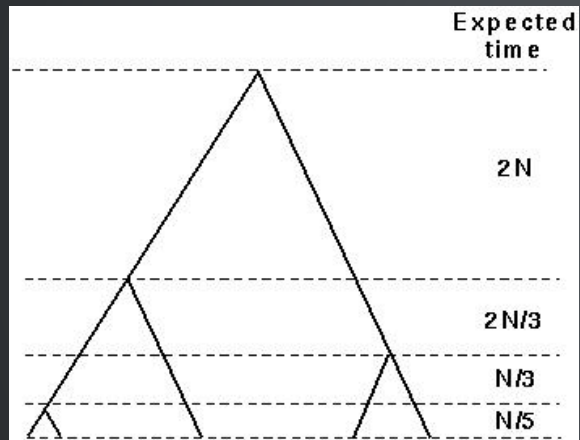
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- The constant rate of substitution for neutral mutations can be used to estimate divergence time.

APPENDIX: SOME MATH BEHIND RANDOM GENETIC DRIFT



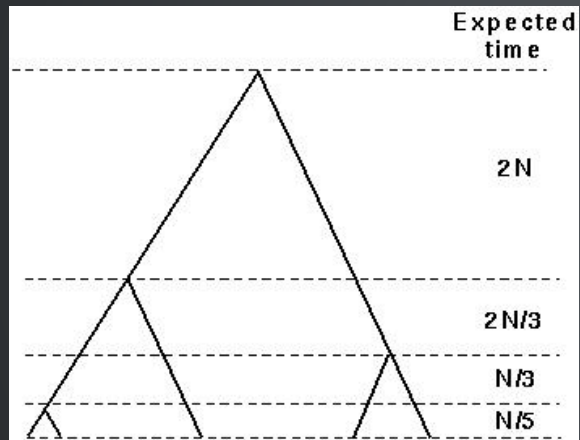
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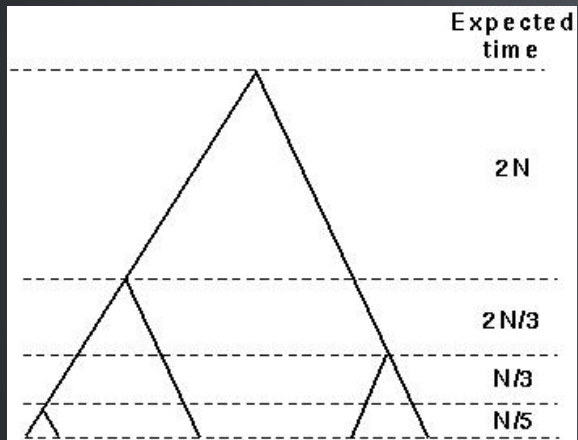
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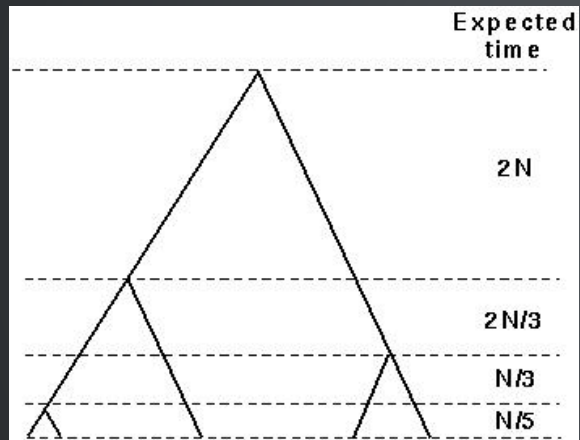
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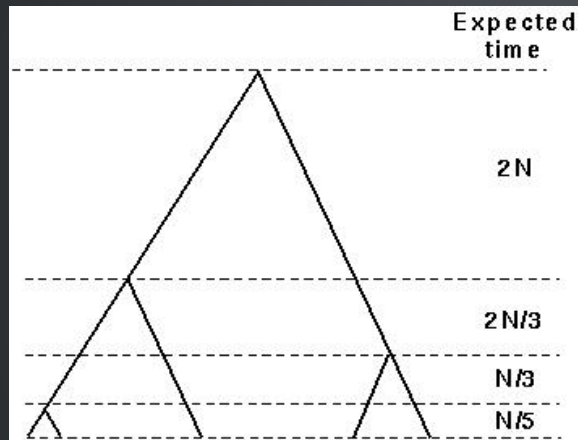
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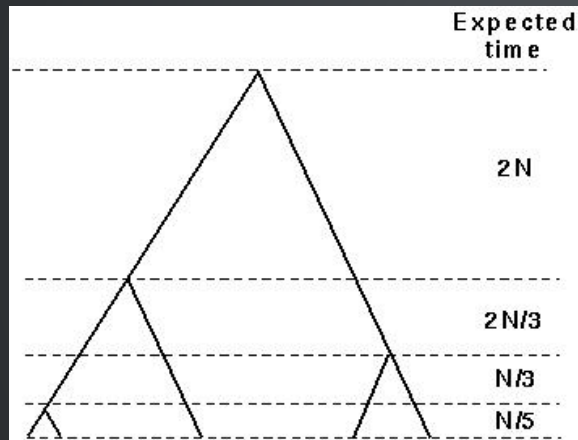
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- $TMCA = \sum_{i=2}^n \frac{2N}{\binom{i}{2}} = 4N \frac{n-1}{n}$

SELECTION

Natural selection (*Darwinian*)

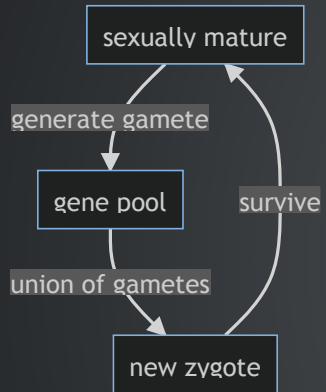
Populations change (*evolve*) over time as the environment (*nature*) favors (*selects*) features that enhance the ability to survive and reproduce.

SELECTION

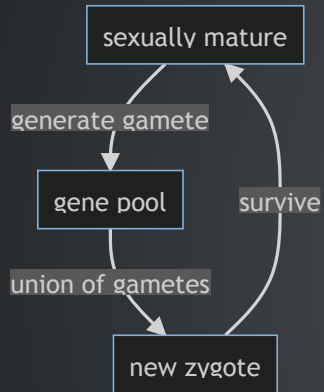
? Question

We often use the phrase "*survival of the fittest*" to describe Darwinian selection. If an individual is physically strong, resistant to disease, and lives a long life, does this individual have a higher fitness?

SELECTION

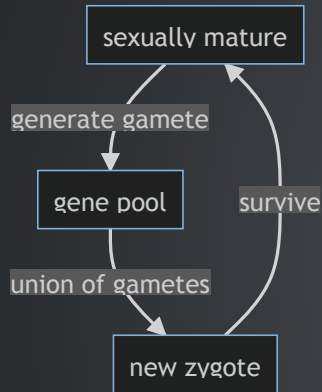


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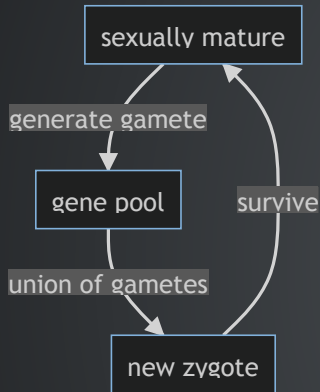
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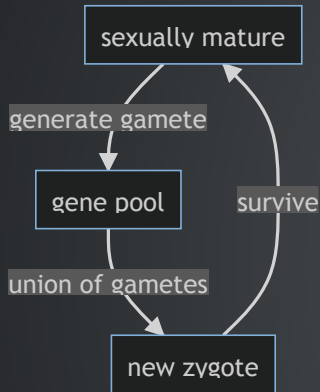
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- Where does selection operate in the left diagram?

HOW NATURAL SELECTION ALTERS ALLELE FREQUENCY?

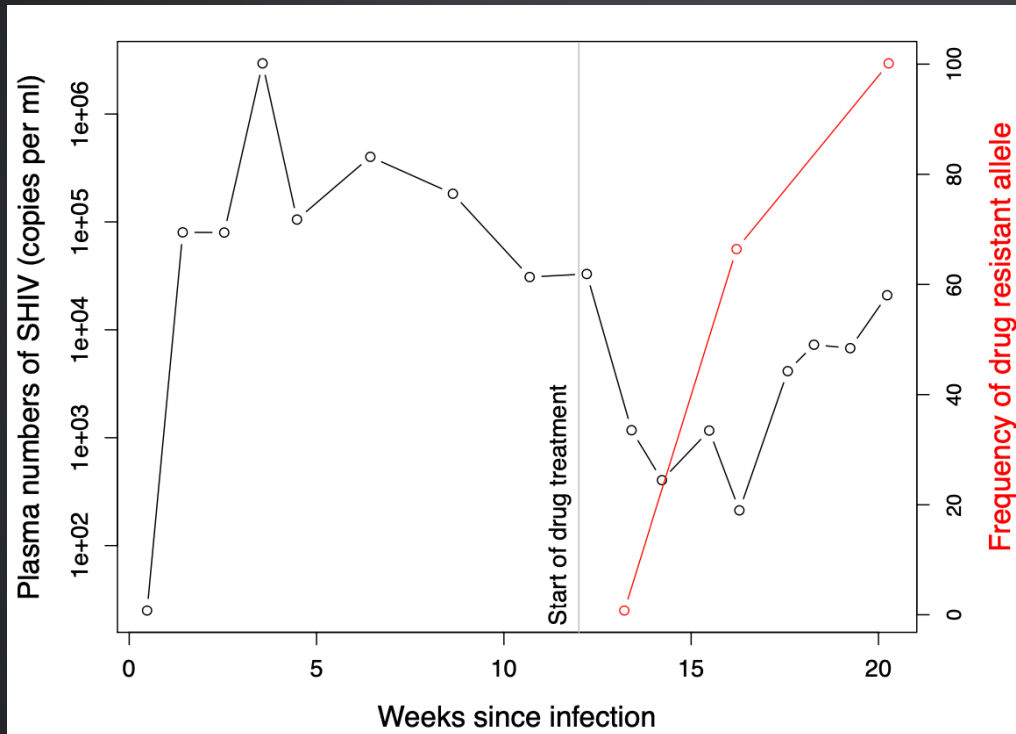


Figure 10.1: The rapid evolution of drug-resistant SHIV. The viral load of SHIV in the blood of a macaque (black line), the frequency of a drug resistance mutation (red line). Data from Feder et al. (2017).

HOW NATURAL SELECTION ALTERS ALLELE FREQUENCY?

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Genotype	A/A	A/a	a/a
Average number of offspring (W)	10	10	5
Relative fitness (w)	1.0	1.0	0.5
Genotype frequency	0.01	0.18	0.81

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Genotype frequency	0.01	0.18	0.81

HOW NATURAL SELECTION ALTERS ALLELE FREQUENCY?

- A is a favored dominant allele

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- The relative contribution of each genotype to the gene pool is determined by **the product of its fitness and frequency**

Genotype	A/A	A/a	a/a	Sum
Relative contribution	1×0.01	1×0.18	0.5×0.81	
	= 0.01	= 0.18	= 0.405	0.595

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 - $P_{AA} = ?$, etc.

HOW NATURAL SELECTION ALTERS ALLELE FREQUENCY?

More generally

define $\bar{w} = p^2 w_{A/A} + 2pq w_{A/a} + q^2 w_{a/a}$ is the mean relative fitness of the population

$$\text{then, } p' = \frac{p^2 w_{A/A} + \frac{1}{2} \times 2pq w_{A/a}}{\bar{w}} = p \frac{p w_{A/A} + q w_{A/a}}{\bar{w}}$$

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Q: the speed of allele frequency change depends on ___?

FORMS OF SELECTION

1. Directional selection

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- Positive selection

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- Purifying (negative) selection

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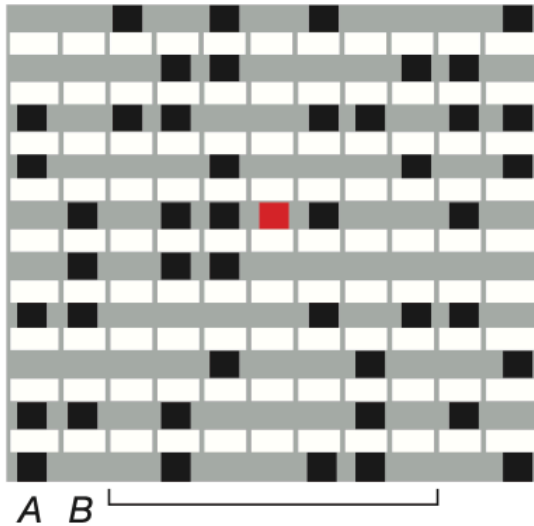
2. Balancing selection

- if the heterozygous class has higher fitness than either homozygous ones, selection would favor the **maintenance of both alleles**.
- can you think of example scenarios for balancing selection?
- are there other types of selection that can **maintain genetic diversity** rather than **eliminating it**?

SIGNATURES OF POSITIVE SELECTION

Positive selection leaves a distinct signature

Haplotypes before selection

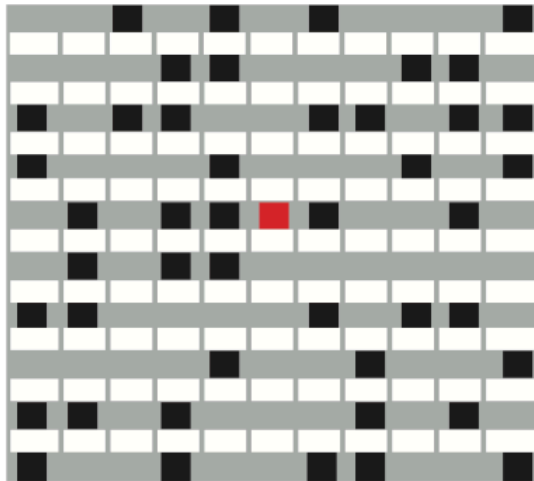


1: Introduction to Genetic Analysis, ed 11, Fig. 18-22

SIGNATURES OF POSITIVE SELECTION

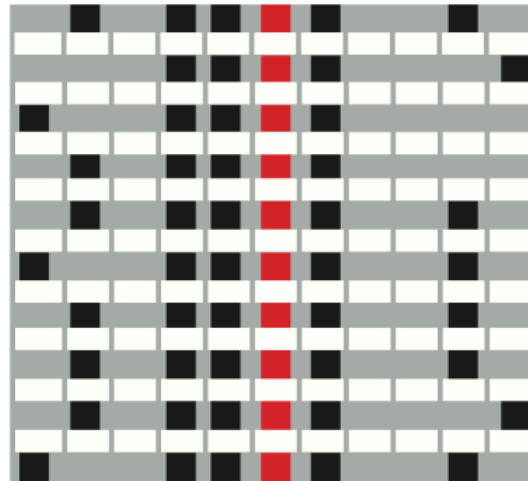
Positive selection leaves a distinct signature

Haplotypes before selection



A B

Haplotypes after selection



A B
Selective sweep

SIGNATURES OF POSITIVE SELECTION

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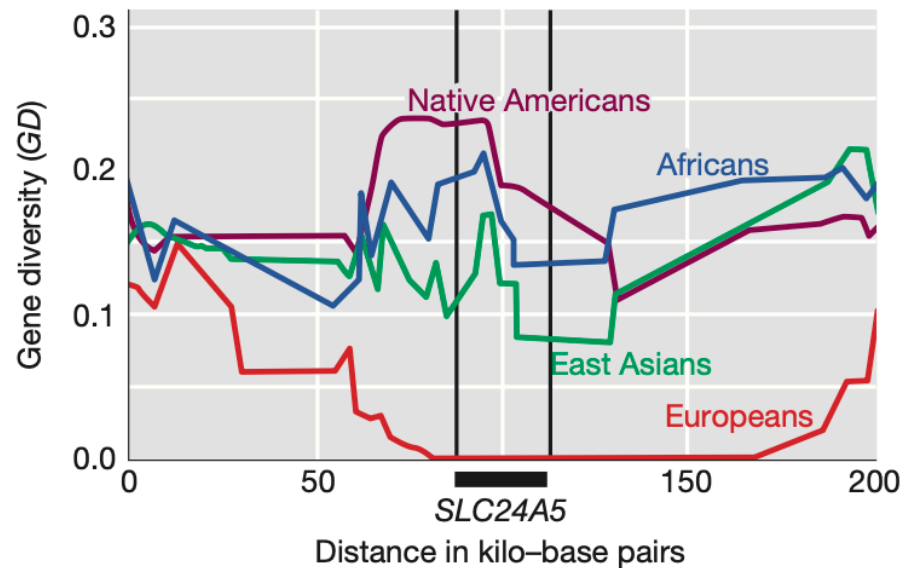
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SIGNATURES OF POSITIVE SELECTION

- By driving the favored allele to fixation "very fast", positive selection causes a selective sweep that ends up
 - also fix linked sites or drive the linked allele to high frequency
 - hence removing genetic variation from the population
 - creates LD
- Note that this signature is **transient**. What does this mean, why?

USE SELECTIVE SWEEP SIGNATURE TO INFER POSITIVE SELECTION

In Europe, a selective sweep caused a loss of all diversity at the *SLC24A5* locus



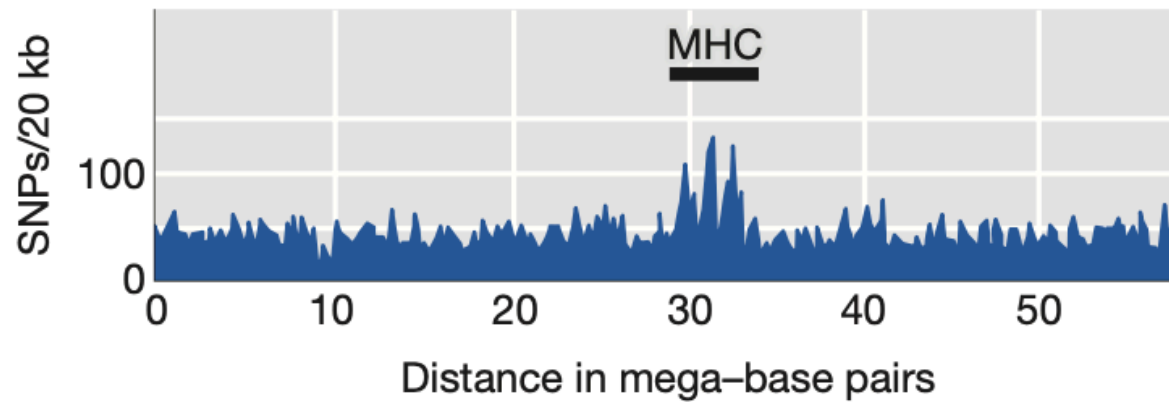
1: Introduction to Genetic Analysis, ed 11, Fig. 18-23

TABLE 8-6 Some Genes Showing Evidence for Natural Selection in Specific Human Populations

Gene	Presumed Trait	Population
<i>EDA2R</i> (ectodysplasin A2 receptor)	Male pattern baldness	Europeans
<i>EDAR</i> (ectodysplasin A receptor)	Hair morphology	East Asians
<i>FY^{null}</i> (Duffy antigen)	Resistance to malaria	Africans
<i>G6PD</i> (glucose-6-phosphate dehydrogenase)	Resistance to malaria	Africans
<i>Hb</i> (hemoglobin B)	Resistance to malaria	Africans
<i>KITLG</i> (KIT ligand)	Skin pigmentation	East Asians and Europeans
<i>LARGE</i> (glycosyltransferase)	Resistance to Lassa fever	Africans
<i>LCT</i> (lactase)	Lactase persistence; ability to digest milk sugar as an adult	Africans, Europeans
<i>LPR</i> (leptin receptor)	Processing of dietary fats	East Asians
<i>MC1R</i> (melanocortin receptor)	Hair and skin pigmentation	East Asians
MHC (major histocompatibility complex)	Infectious disease resistance	Multiple populations
<i>OCA2</i> (oculocutaneous albinism)	Skin pigmentation and eye color	Europeans
<i>PPARD</i> (peroxisome proliferator-activated receptor delta)	Processing of dietary fats	Europeans
<i>SI</i> (sucrase-isomaltase)	Sucrose metabolism	East Asians
<i>SLC24A5</i> (solute carrier family 24)	Skin pigmentation	Europeans and West Asians
<i>TYRP1</i> (tyrosinase-related protein 1)	Skin pigmentation	Europeans

Source: P. C. Sabeti et al., *Science* 312, 2006, 1614–1620; P. C. Sabeti et al., *Nature* 449, 2007, 913–919; B. F. Voight et al., *PLoS Biology* 4, 2006, 446–458; J. K. Pickrell et al., *Genome Research* 19, 2009, 826–837.

Balancing selection can lead to regions of unusually high genetic diversity



1: Introduction to Genetic Analysis, ed 11, Fig. 18-24

